MOTOR VIBRATION EXPERIMENTAL DATA ANALYSIS

Dataset	adxl345_5sec_calibrated
Sensor	ADXL345
Controller	ESP32
Sampling Frequency	1559.00 Hz
Number of Samples	49939
Duration	32.033 s

1. Dataset Information

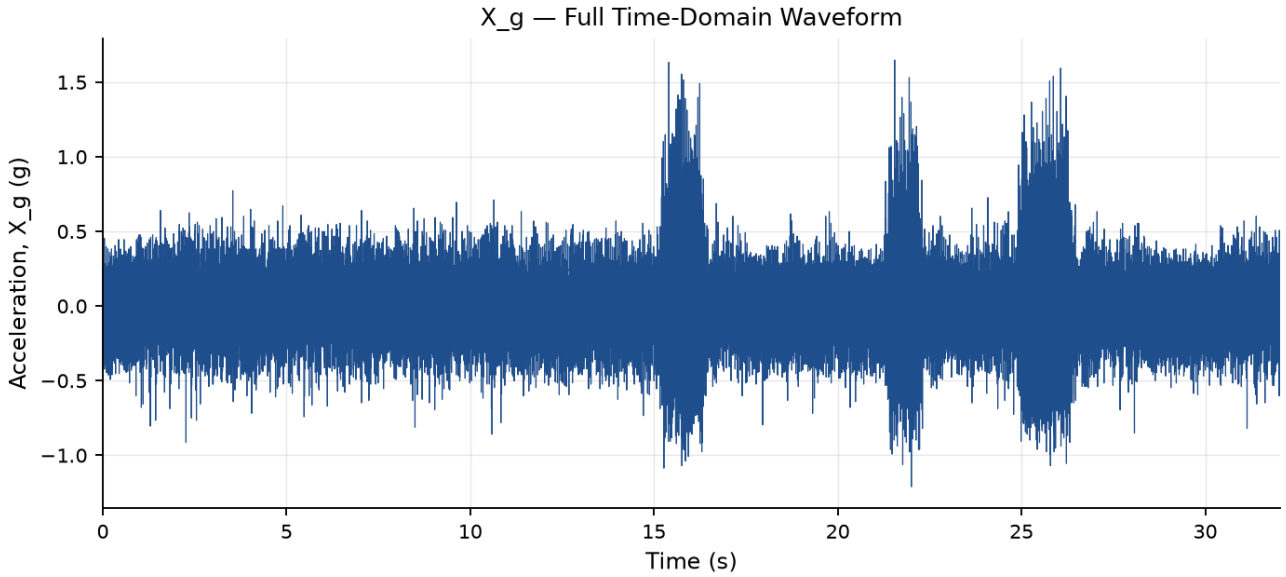
Parameter	Value
Input file	adxl345_5sec_calibrated.csv
Samples	49939
Sampling frequency	1559.00 Hz
Duration	32.033 s
Nyquist frequency	779.50 Hz
Frequency resolution	0.0312 Hz
Sensor	ADXL345
Controller	ESP32
Axes	X, Y, Z
Units	g (gravitational acceleration)

2. Signal Processing Methodology

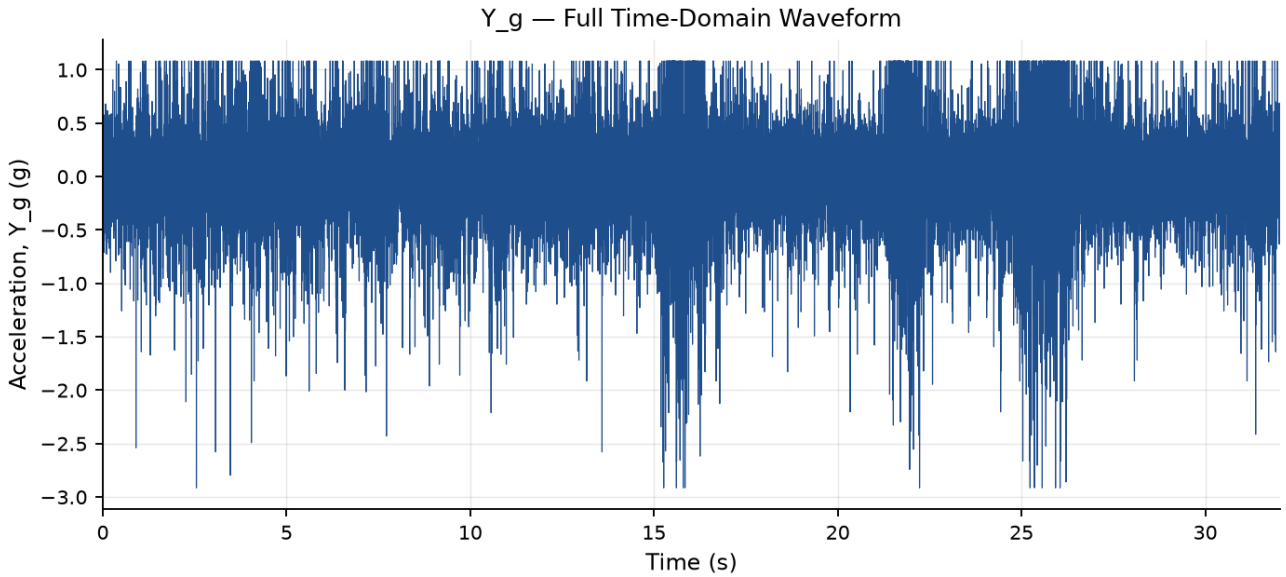
Vibration data were acquired with an ADXL345 accelerometer connected to an ESP32, sampled at 1559.0 Hz. The input CSV is already calibrated/processed and contains acceleration channels X_g, Y_g, Z_g. The mean (DC) component of each channel was removed to obtain AC vibration signals X_ac, Y_ac, Z_ac. A resultant vibration signal was formed as $R_g = \sqrt{X_{ac}^2 + Y_{ac}^2 + Z_{ac}^2}$. Frequency-domain analysis applies a Hann window before computing a single-sided amplitude-spectrum FFT (window-gain corrected) over the complete 0–Nyquist range. Power Spectral Density is estimated with Welch's method, and time–frequency content is examined with a short-time Fourier spectrogram. Significant spectral peaks are detected with an adaptive prominence threshold derived from each dataset's own robust noise floor (median and median-absolute-deviation of the power spectrum), so the detection automatically scales with the amplitude of the current dataset rather than relying on a fixed value. All numerical results below are directly measured quantities; the report performs no automatic engineering interpretation or fault classification.

3. Time-Domain Waveforms (Processed Signals)

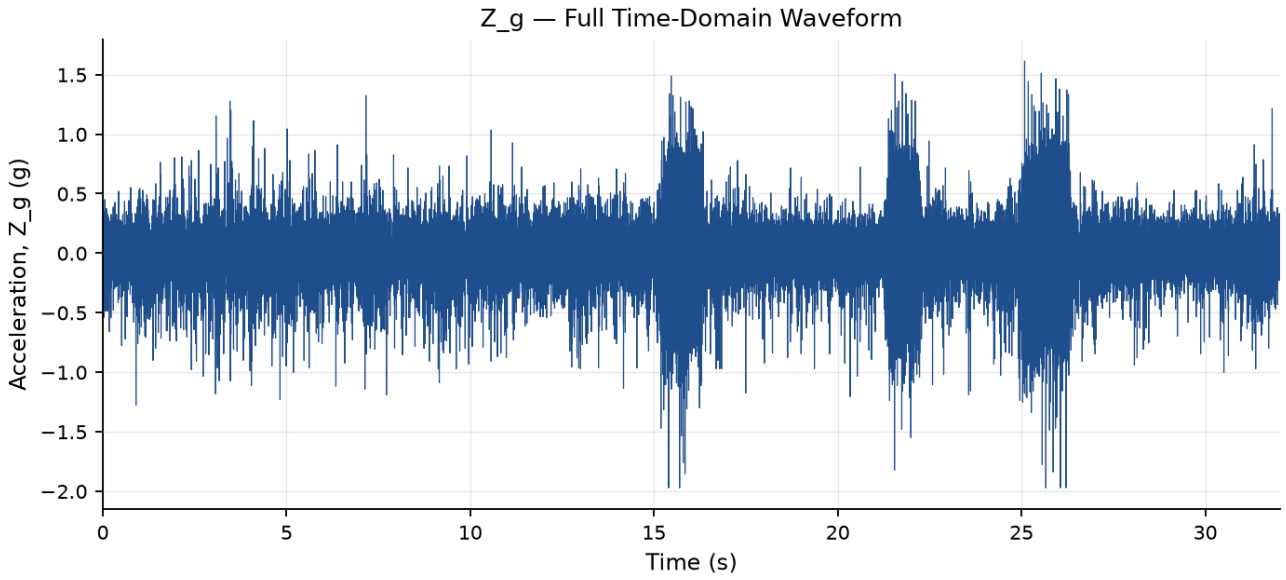
Full-duration waveforms of the processed acceleration channels, followed by a 3-second zoom window for waveform-shape inspection.



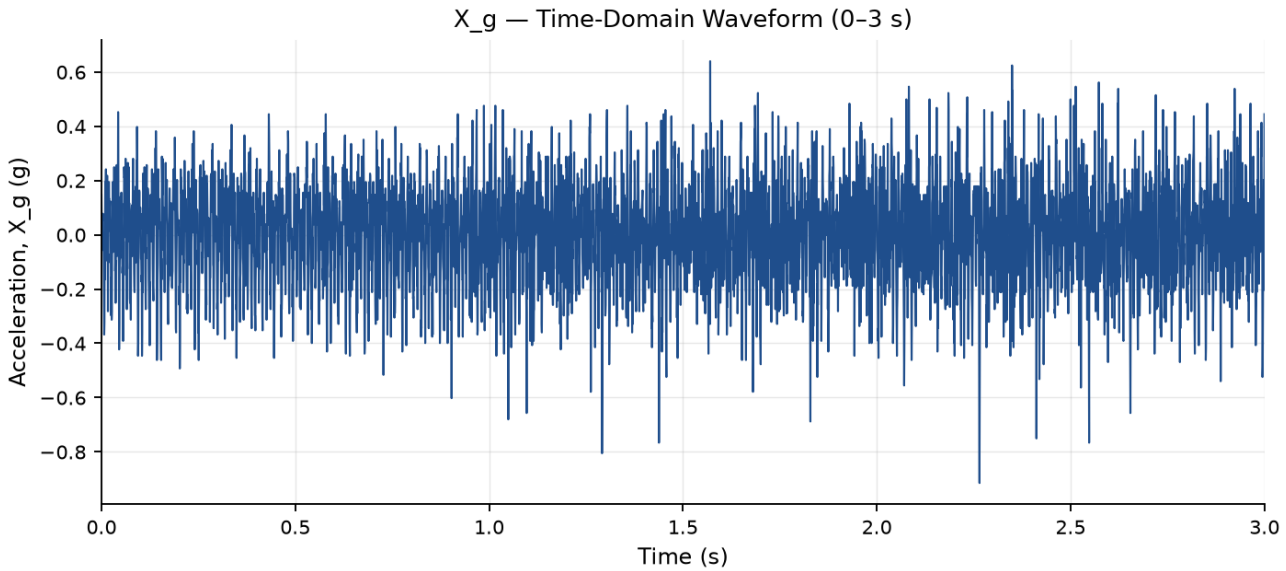
X_g — full time-domain waveform.



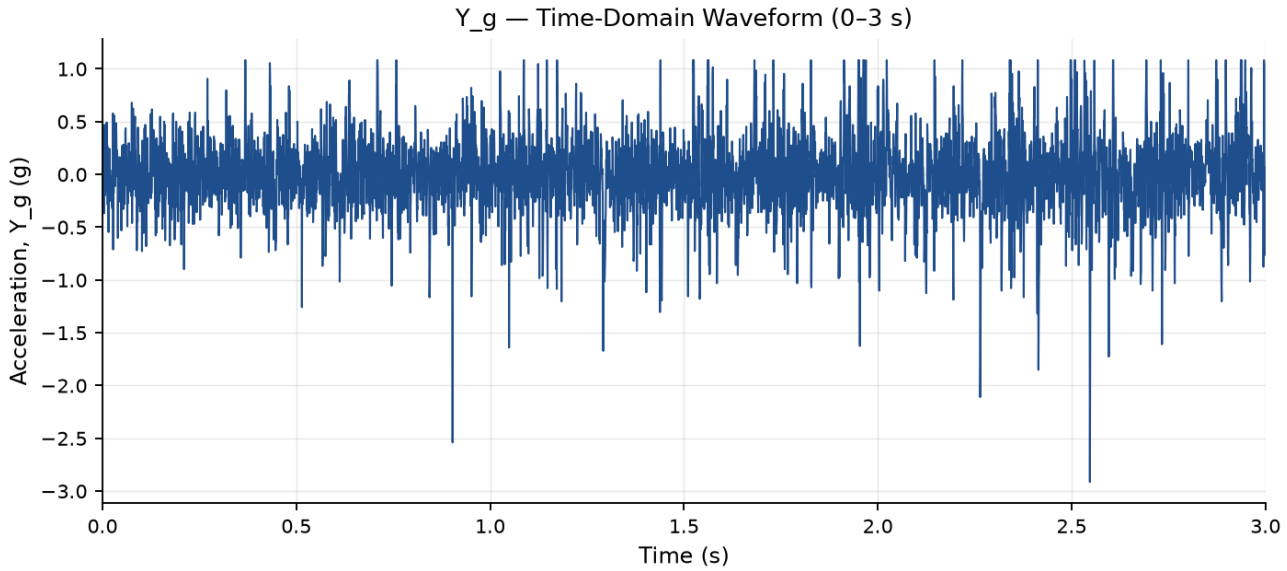
Y_g — full time-domain waveform.



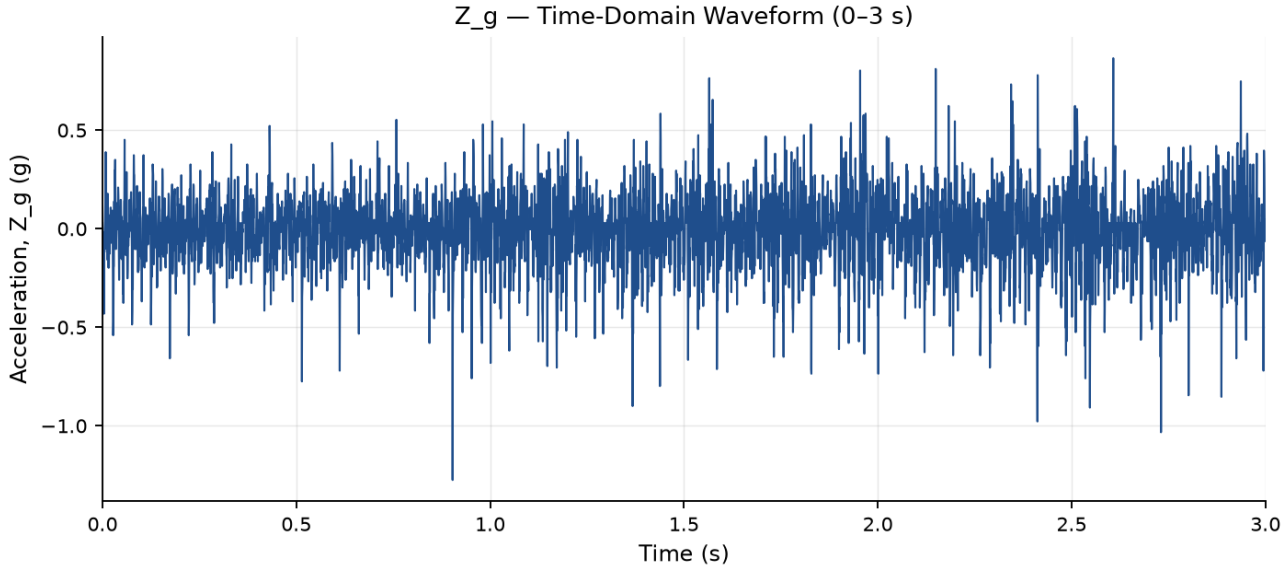
Z_g — full time-domain waveform.



X_g — zoomed waveform (0–3 s).



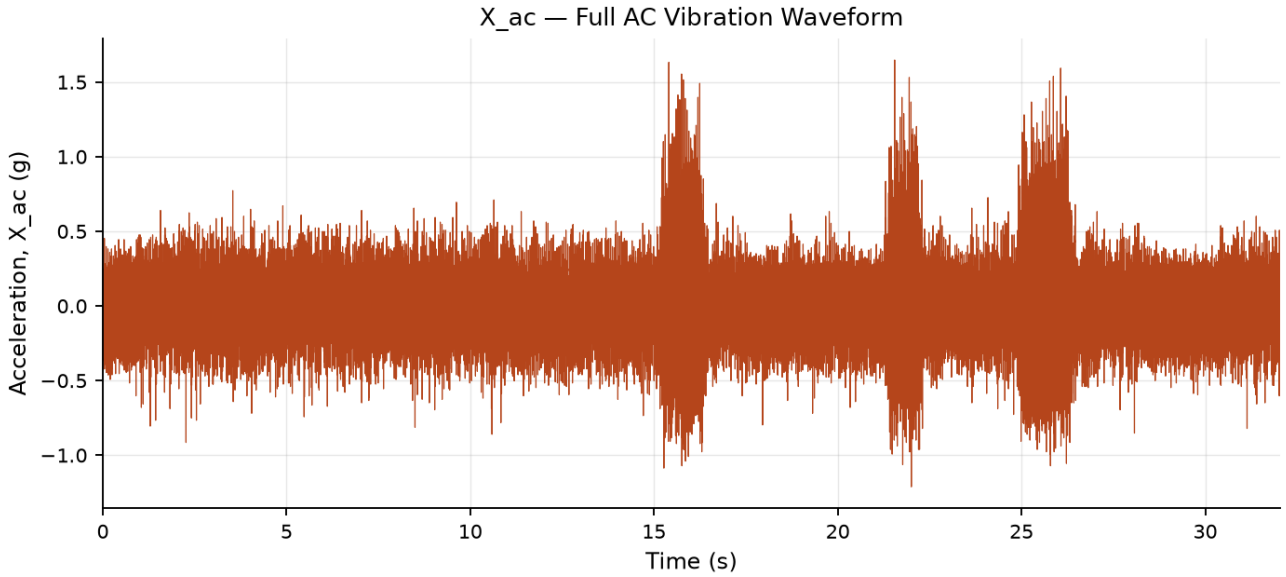
Y_g — zoomed waveform (0–3 s).



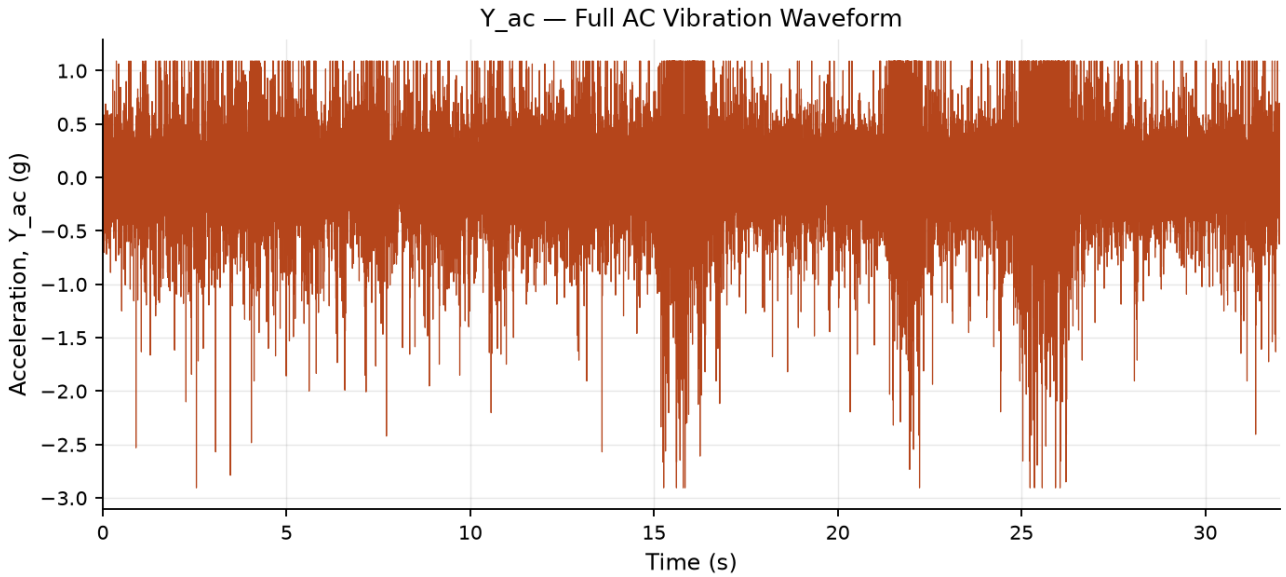
Z_g — zoomed waveform (0–3 s).

4. AC Vibration Waveforms

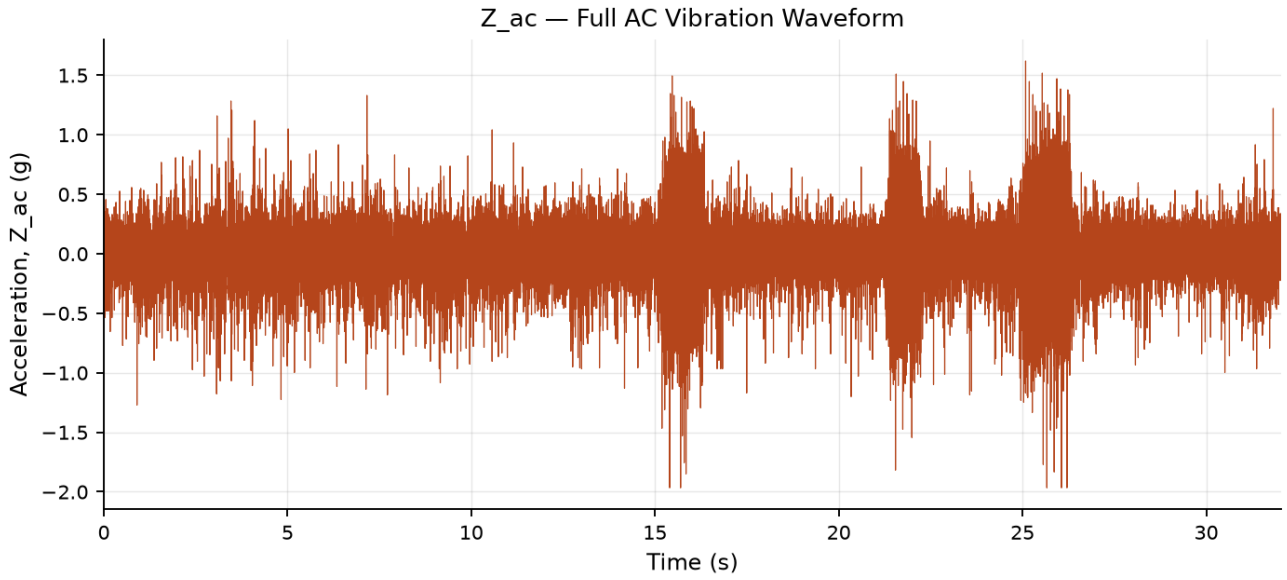
DC-removed AC vibration signals (X_ac, Y_ac, Z_ac) and the resultant vibration signal R_g = sqrt(X_ac²+Y_ac²+Z_ac²). Full duration and zoomed views.



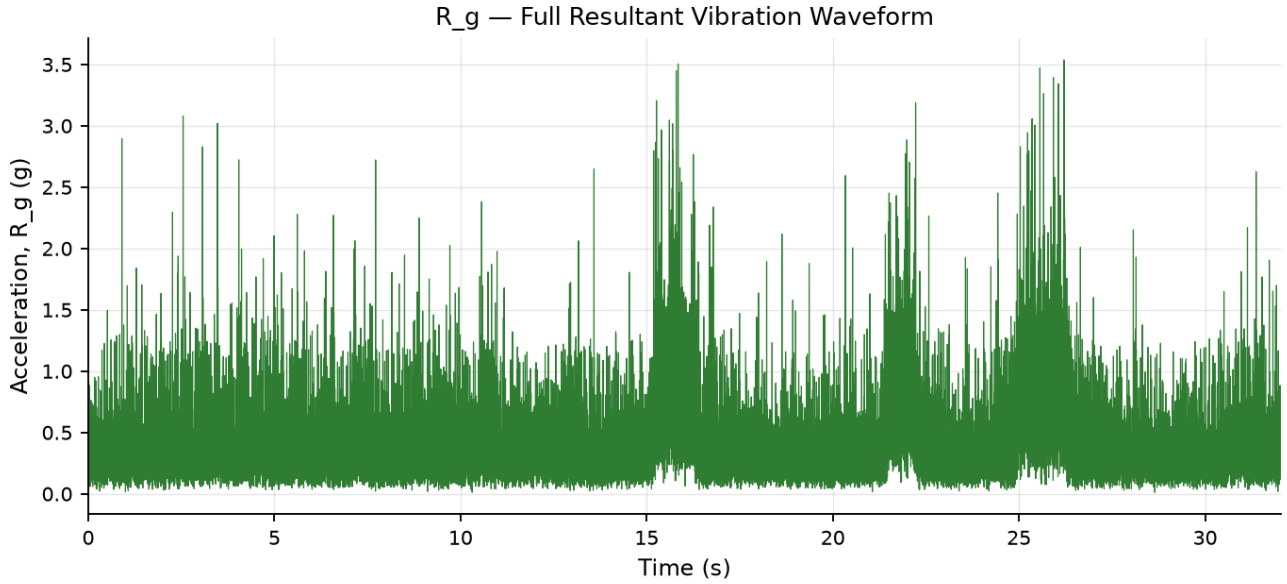
X_ac — full waveform.



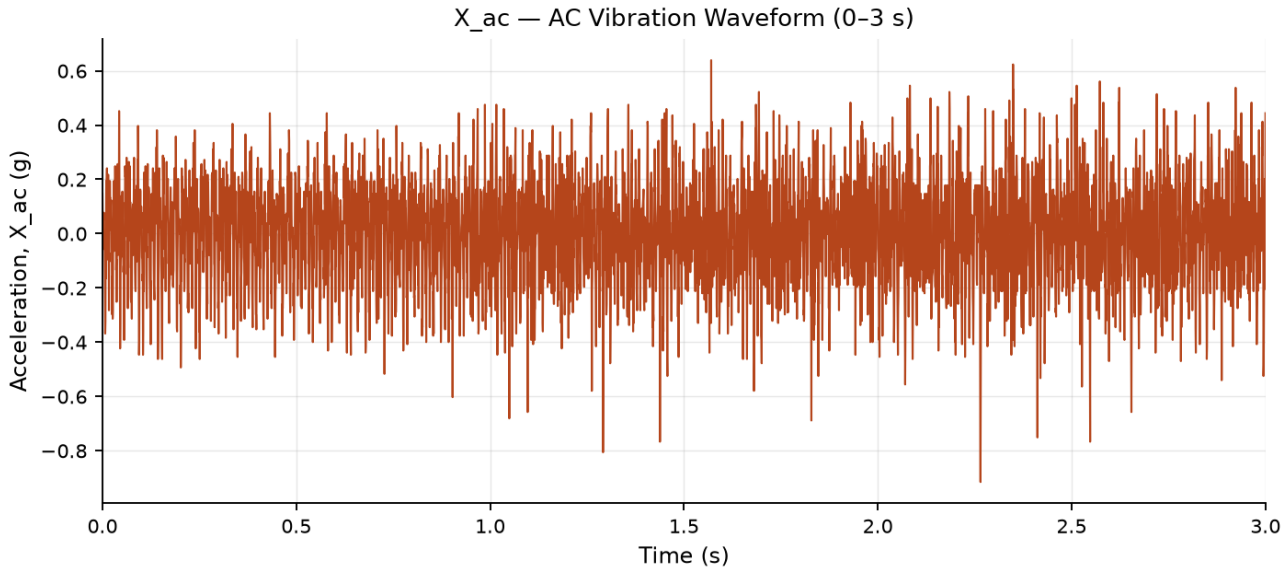
Y_ac — full waveform.



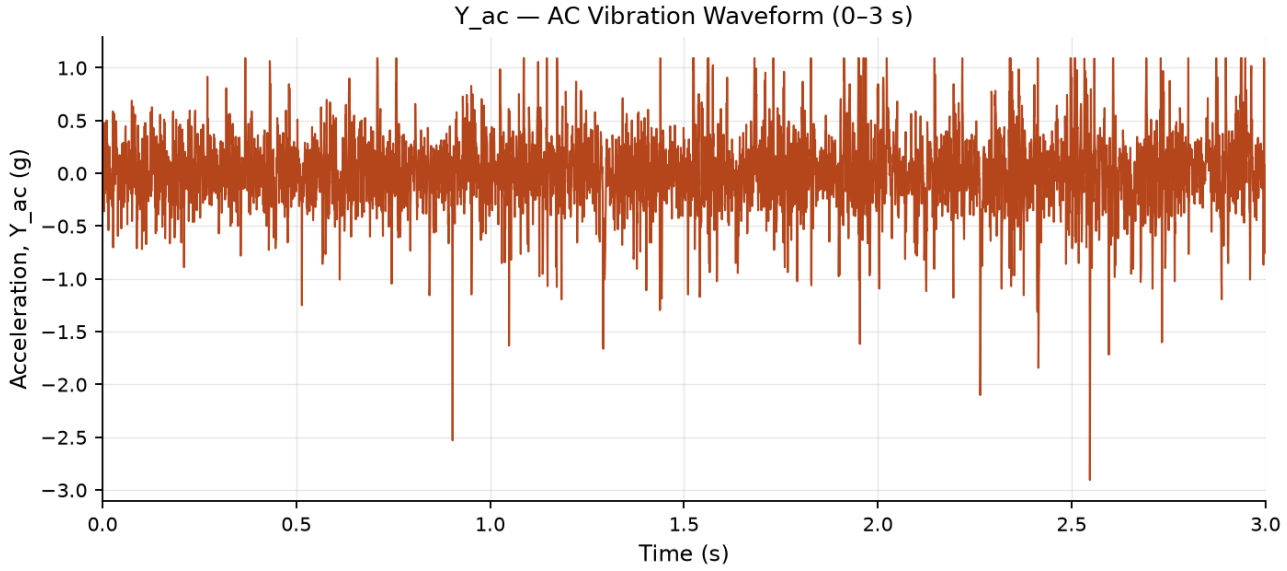
Z_ac — full waveform.



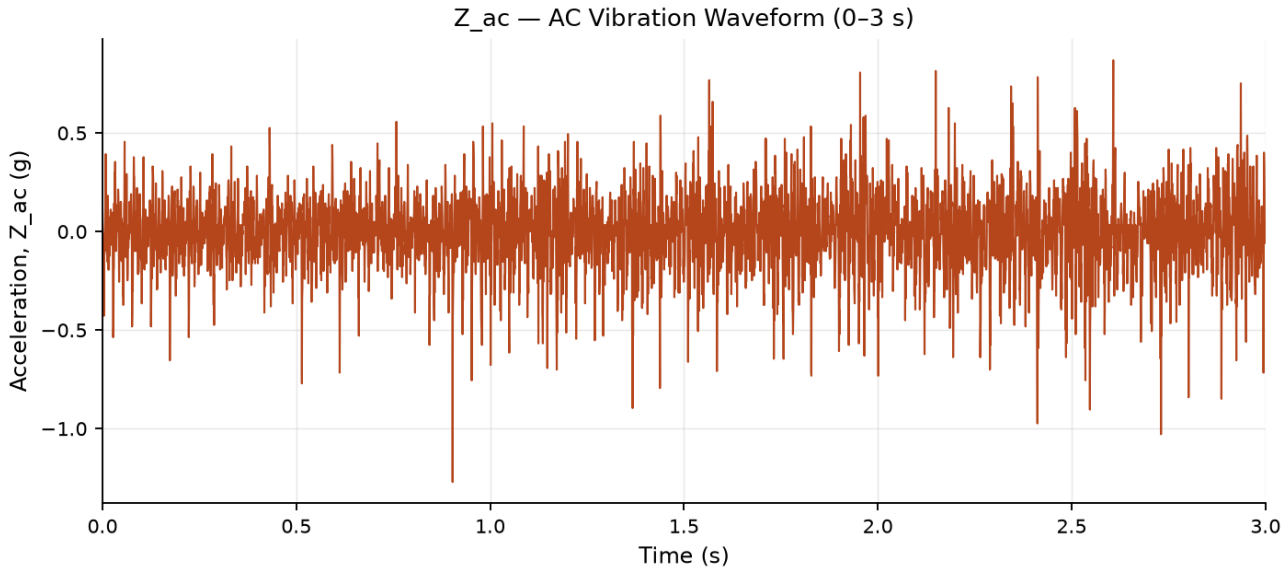
R_g — full waveform.



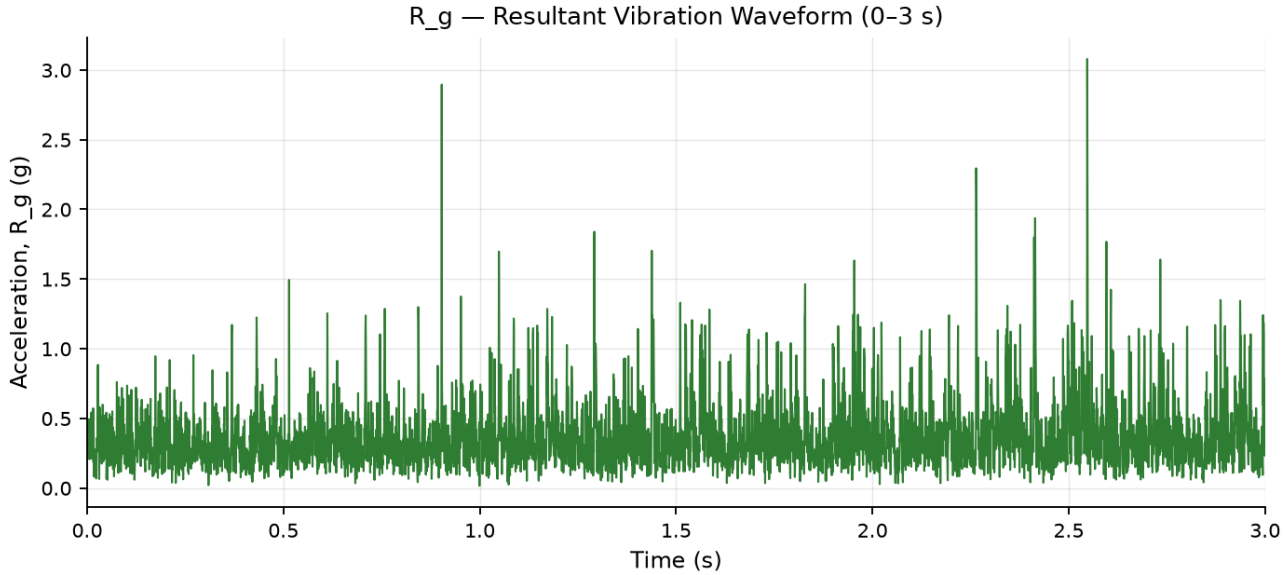
X_ac — zoomed waveform (0–3 s).



Y_ac — zoomed waveform (0–3 s).



Z_ac — zoomed waveform (0–3 s).



R_g — zoomed waveform (0–3 s).

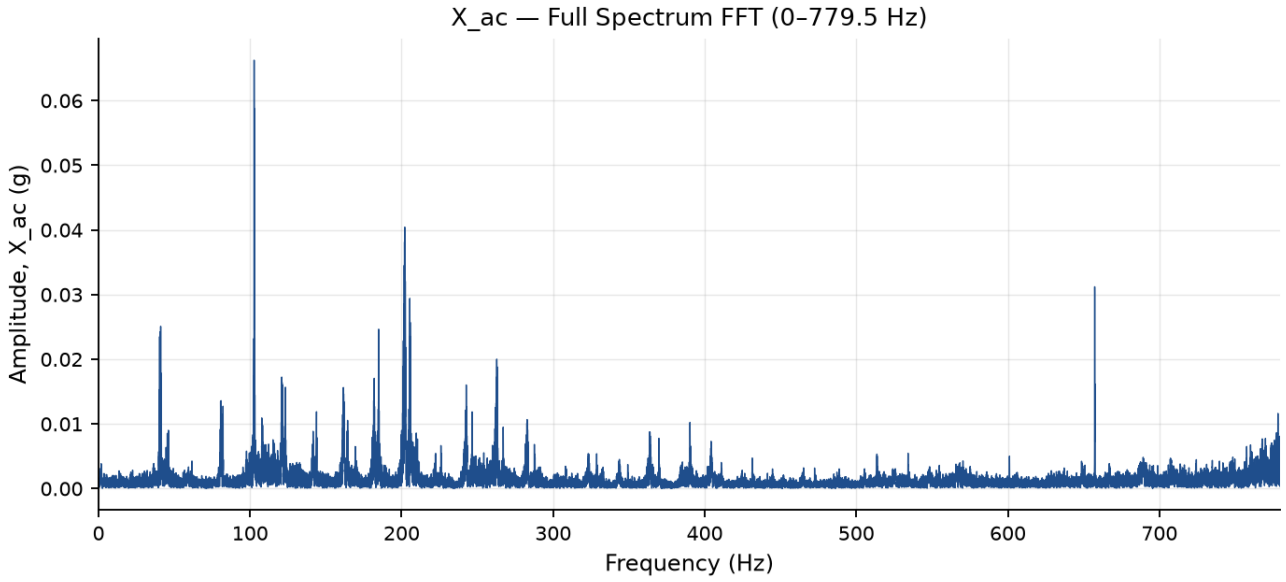
5. Time-Domain Feature Results

Time-domain statistical and shape features computed on X_ac, Y_ac, Z_ac and R_g. Each value is reported once and is not repeated elsewhere in this report.

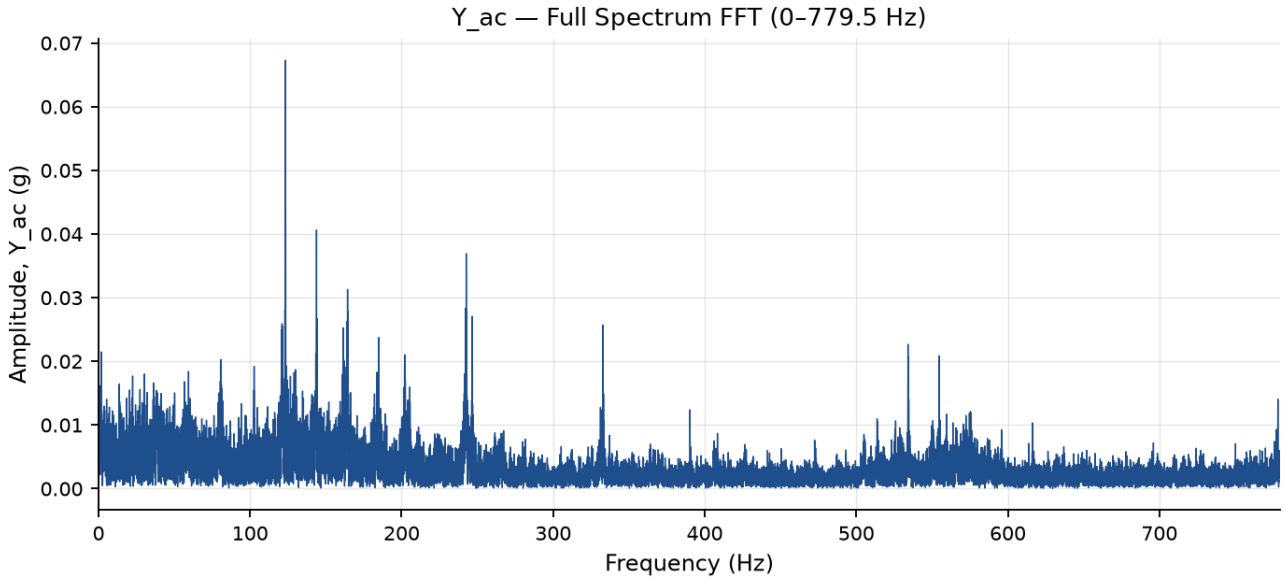
Feature	X_ac	Y_ac	Z_ac	R_g
Mean	6.0114e-18	-1.2094e-18	6.9718e-18	0.42354
Median	0.0063477	0.0082643	0.033065	0.34265
Std Dev	0.23134	0.40203	0.23844	0.30433
Variance	0.053518	0.16163	0.056855	0.092614
RMS	0.23134	0.40203	0.23844	0.52154
Maximum	1.647	1.0903	1.619	3.5363
Minimum	-1.2124	-2.9058	-1.9669	0.010578
Absolute Peak	1.647	2.9058	1.9669	3.5363
Peak to Peak	2.8594	3.9961	3.5859	3.5257
Crest Factor	7.1193	7.2279	8.2491	6.7805
Shape Factor	1.3296	1.4091	1.4388	1.2314
Impulse Factor	9.466	10.185	11.868	8.3493
Clearance Factor	11.421	12.692	14.842	9.2603
Skewness	0.18401	-0.62647	-0.36539	2.3688
Kurtosis	2.3787	4.2229	4.8317	9.3839
Mean Absolute Value	0.17399	0.28531	0.16573	0.42354
Energy	2672.6	8071.4	2839.3	13583
Signal Power	0.053518	0.16163	0.056855	0.272
Range	2.8594	3.9961	3.5859	3.5257
MAD	0.14062	0.20312	0.11719	0.13106
Q1	-0.14209	-0.19486	-0.10756	0.22968
Q3	0.13916	0.21139	0.119	0.51134
IQR	0.28125	0.40625	0.22656	0.28165
Zero Crossing Rate	0.32947	0.34102	0.32993	0

6. Frequency-Domain Analysis (FFT)

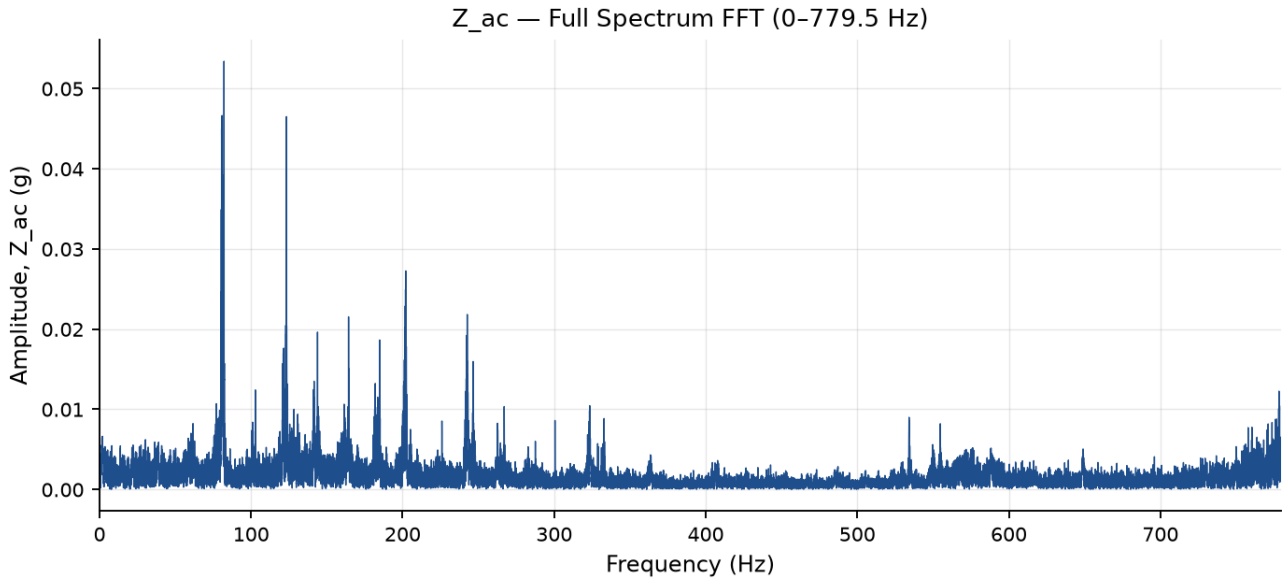
Single-sided amplitude spectra (Hann-windowed) over the complete measured range, 0–779.5 Hz. The 100 Hz zoom below is a visualization aid only and is not an analysis boundary; full-spectrum peak detection was performed over the entire range (see Section 13).



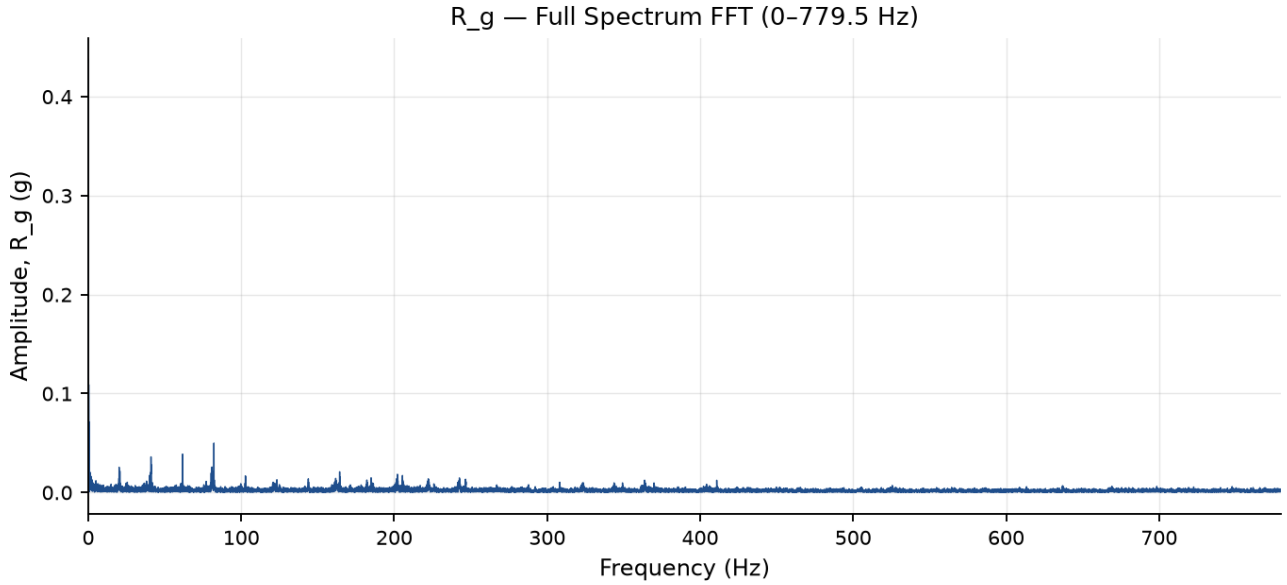
X_ac — full spectrum FFT, 0–779.5 Hz.



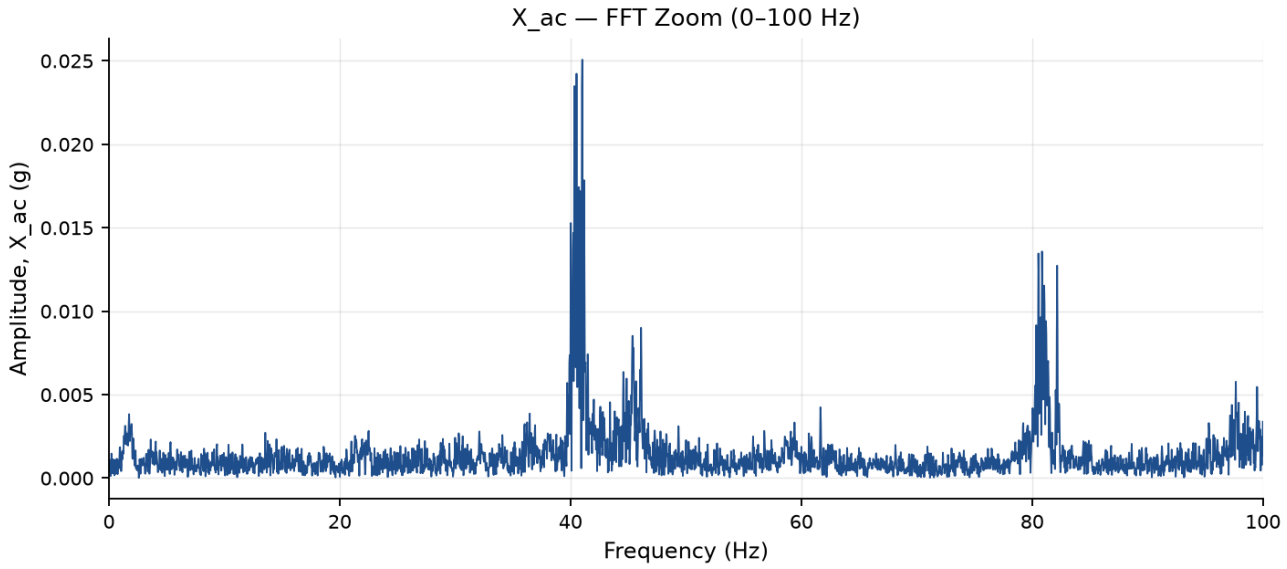
Y_ac — full spectrum FFT, 0–779.5 Hz.



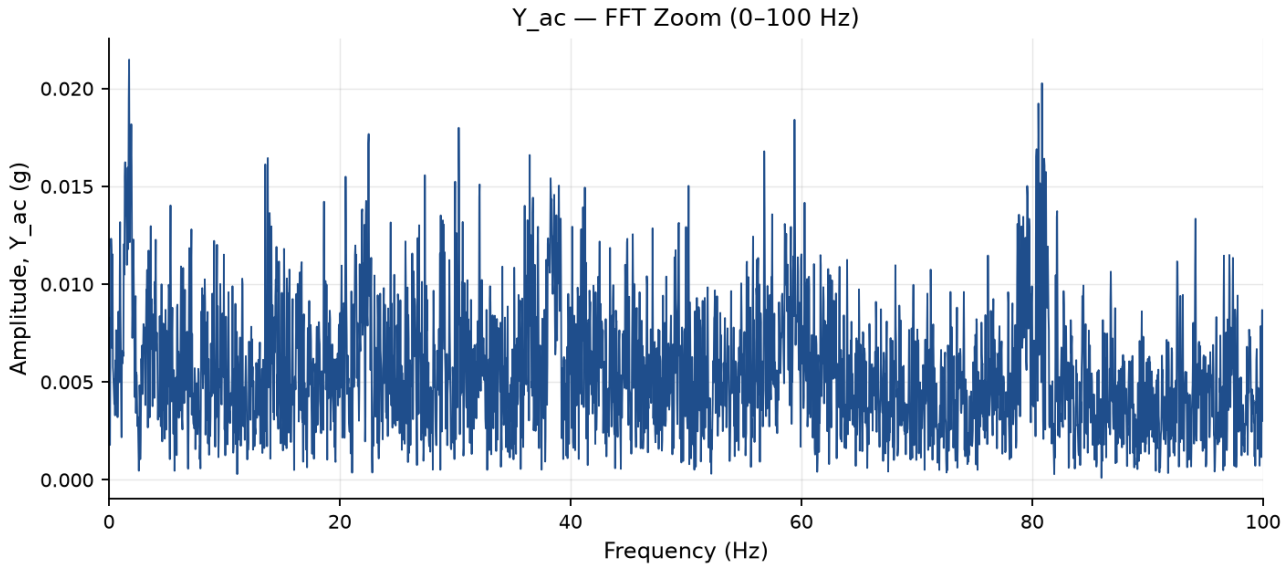
Z_ac — full spectrum FFT, 0–779.5 Hz.



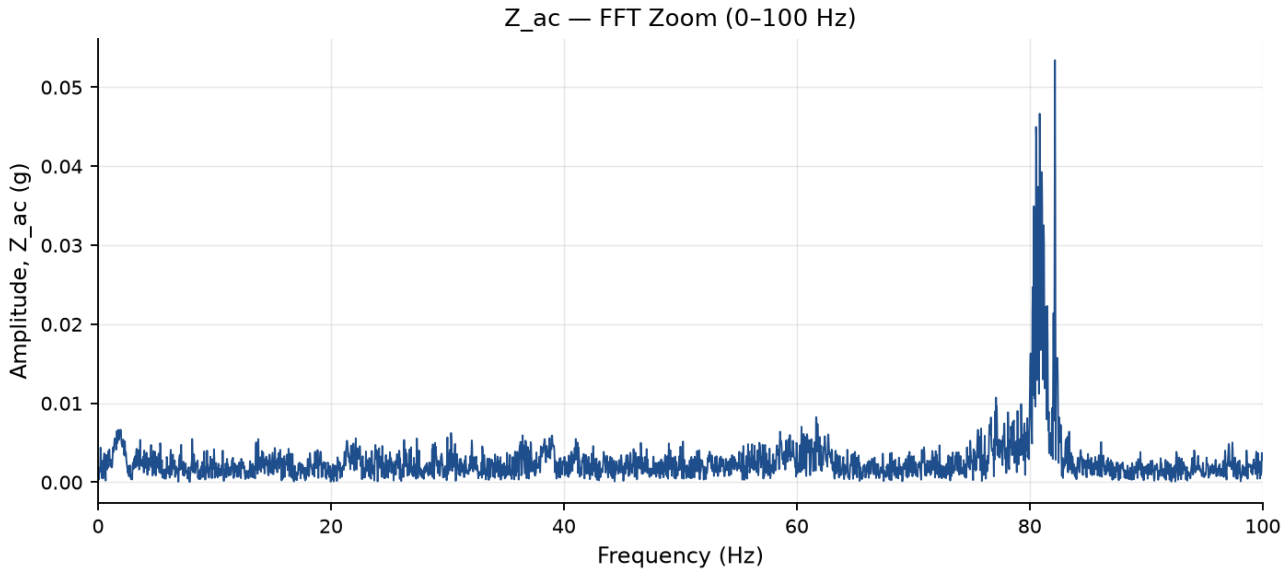
R_g — full spectrum FFT, 0–779.5 Hz.



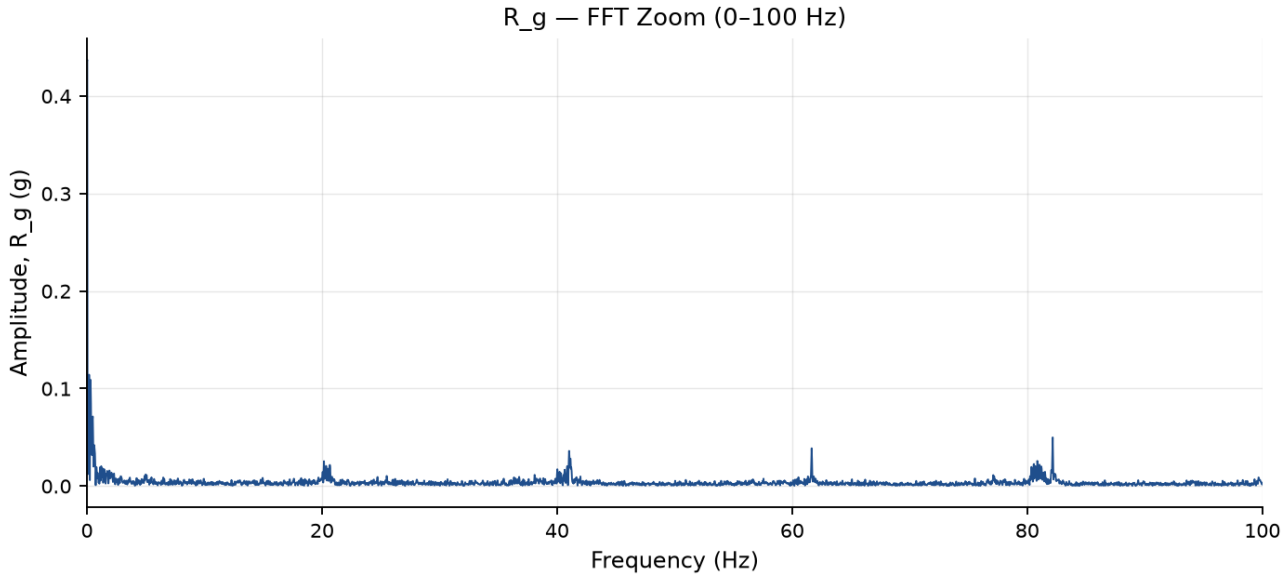
X_ac — FFT, 0–100 Hz zoom (visualization only).



Y_ac — FFT, 0–100 Hz zoom (visualization only).



Z_ac — FFT, 0–100 Hz zoom (visualization only).



R_g — FFT, 0–100 Hz zoom (visualization only).

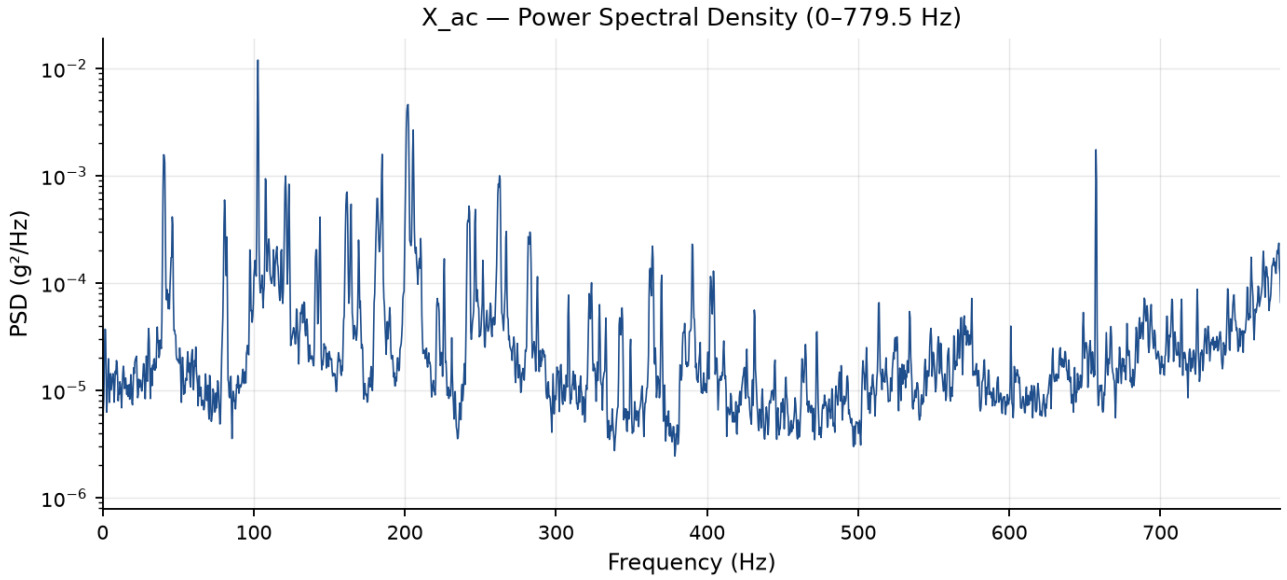
7. Frequency-Domain Feature Results

Spectral features computed from the full 0–Nyquist amplitude/power spectrum (DC excluded). Reported once; not repeated elsewhere in this report.

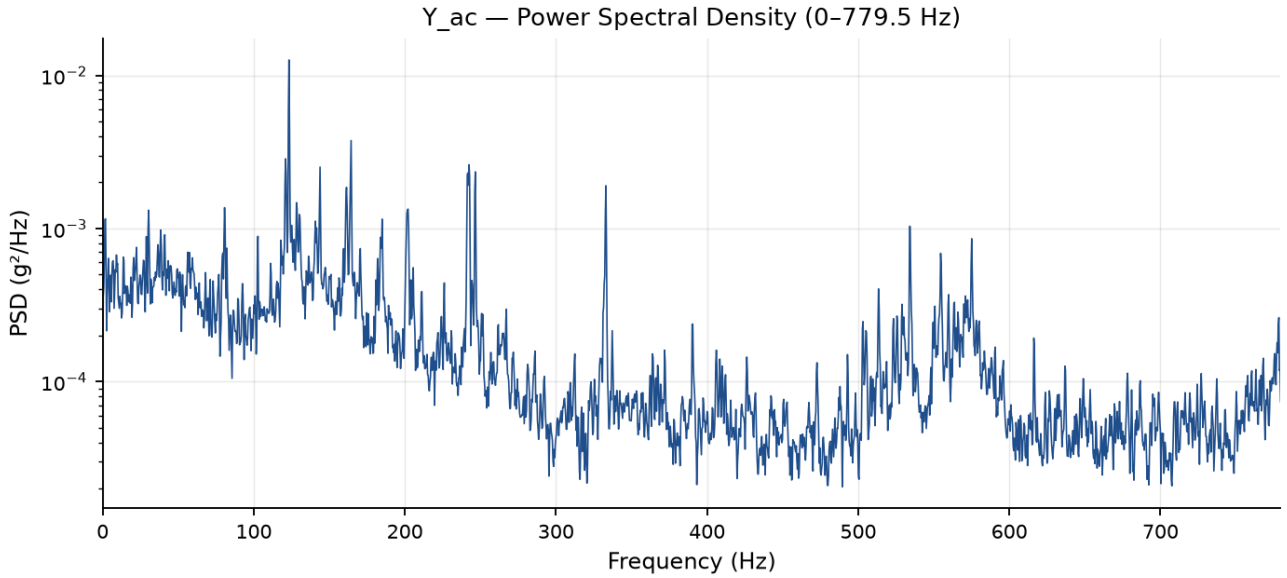
Feature	X_ac	Y_ac	Z_ac	R_g
Dominant Frequency Hz	102.68	123.34	82.166	0.031218
Dominant Amplitude g	0.06623	0.067279	0.053411	0.43705
Spectral Centroid Hz	255.18	225.22	229.84	106.68
Spectral RMS Frequency Hz	325.65	299.05	310.53	213.61
Spectral Bandwidth Hz	202.31	196.73	208.81	185.06
Spectral Energy	0.17885	0.50403	0.18771	0.49178
Spectral Variance	40930	38702	43601	34248
Spectral Skewness	1.4456	1.1816	1.4918	1.9369
Spectral Kurtosis	3.9507	3.3399	4.0213	5.9172
Spectral Flatness	0.1513	0.29934	0.21087	0.15891
Spectral Entropy	0.75468	0.88338	0.80276	0.53352
Spectral Rolloff Hz	514.47	520.87	531.02	272.81
Spectral Peak Power	0.0043864	0.0045265	0.0028527	0.19101
Number of Significant Peaks	5	5	5	4

8. Power Spectral Density

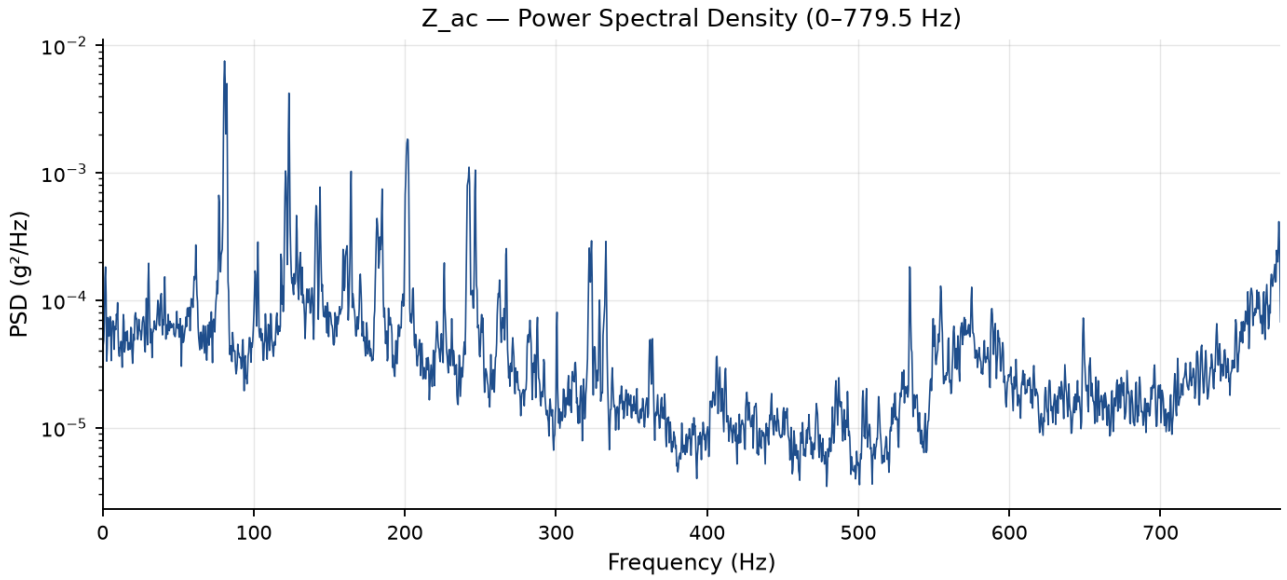
Welch PSD (window=hann, nperseg=4096, noverlap=2048) over the full valid range, followed by a 100 Hz visualization zoom. The full PSD is the primary complete-spectrum analysis.



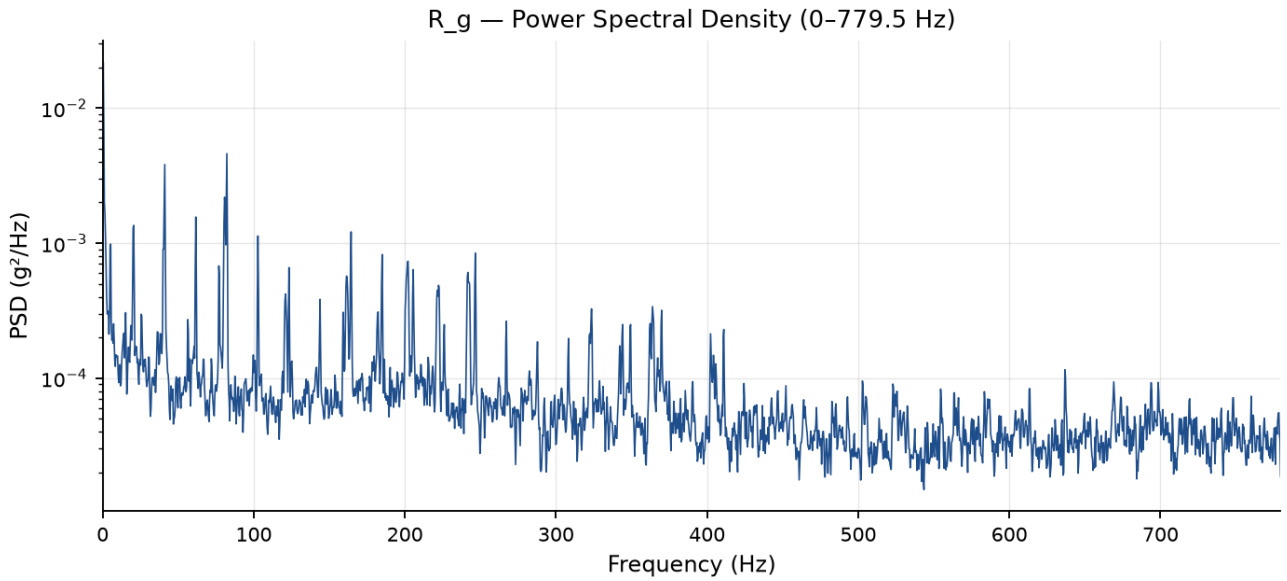
X_ac — full PSD, 0–779.5 Hz.



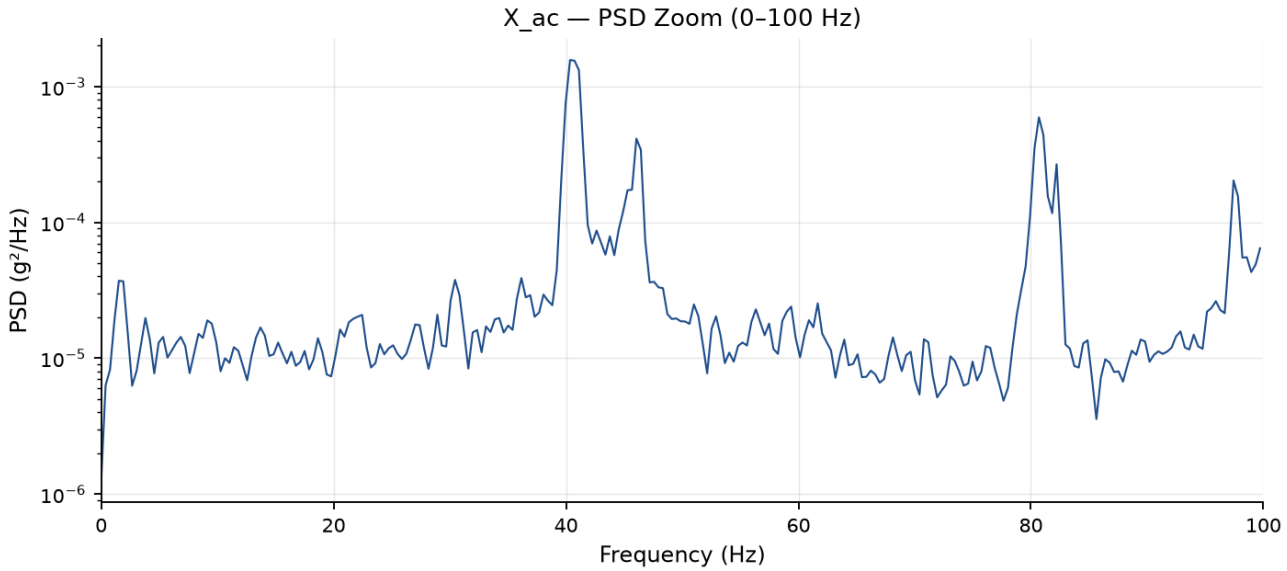
Y_ac — full PSD, 0–779.5 Hz.



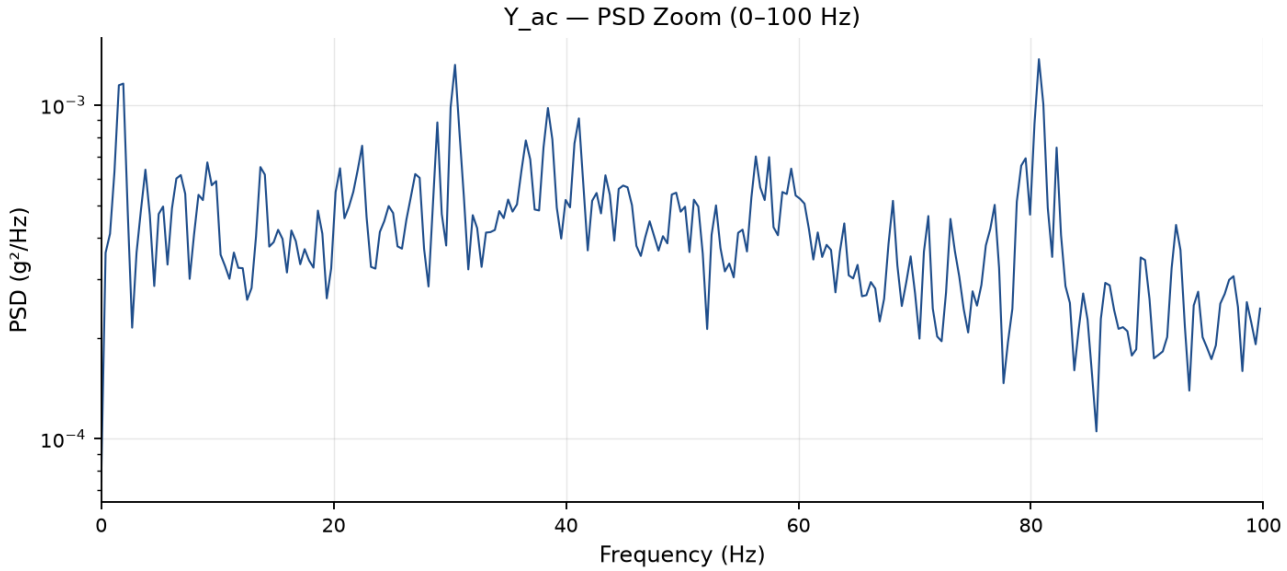
Z_ac — full PSD, 0–779.5 Hz.



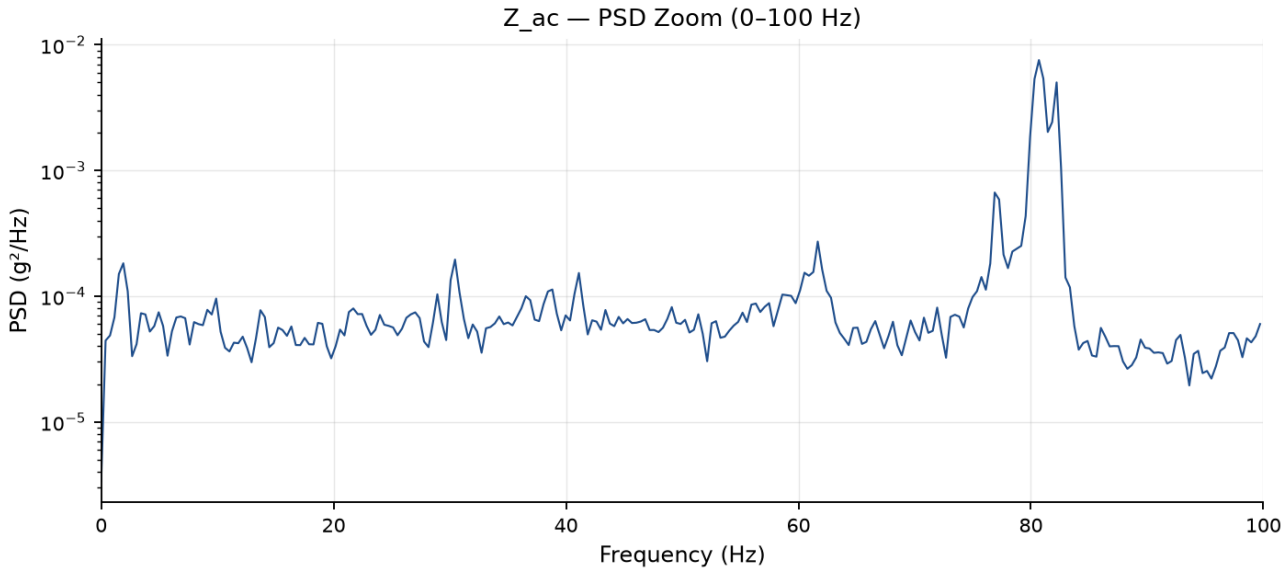
R_g — full PSD, 0–779.5 Hz.



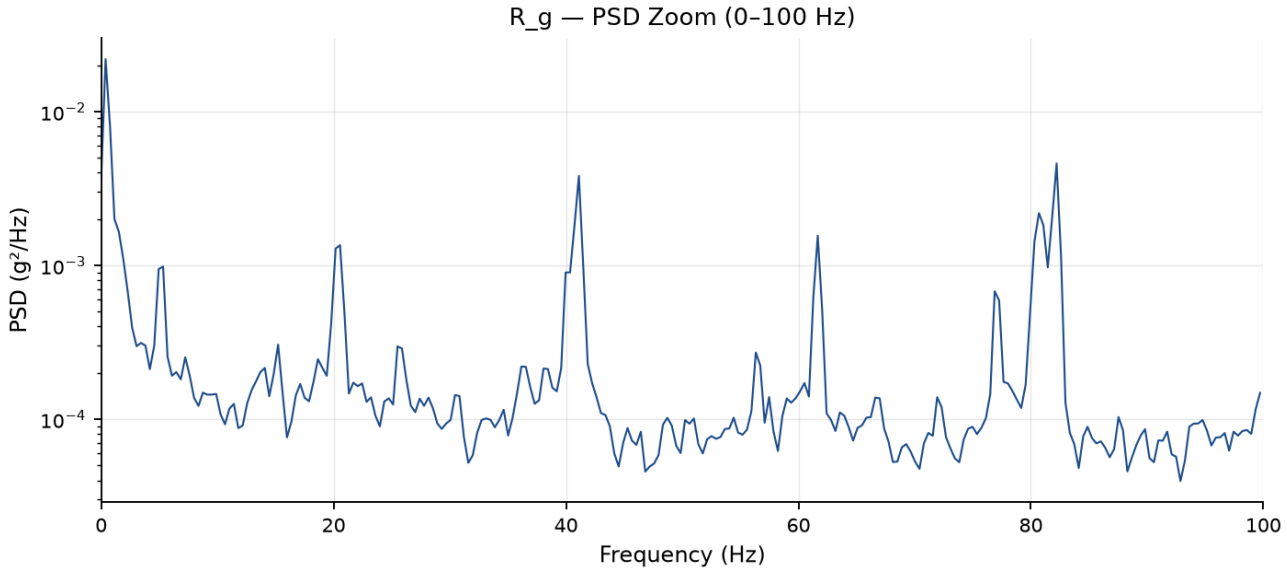
X_ac — PSD, 0–100 Hz zoom.



Y_ac — PSD, 0–100 Hz zoom.



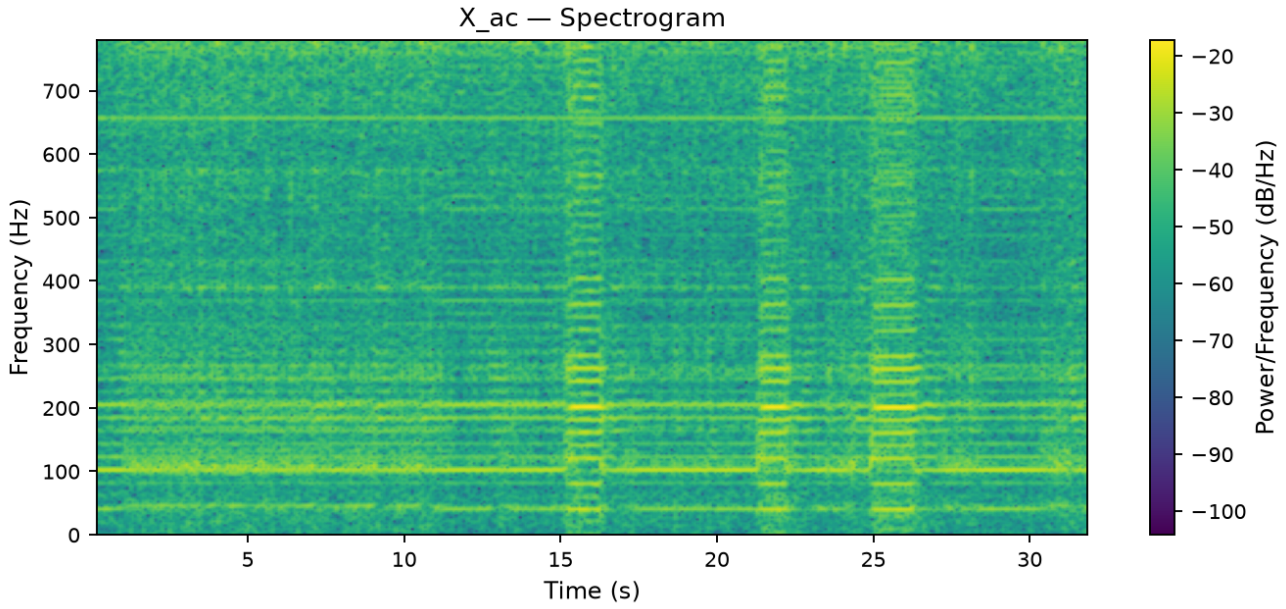
Z_ac — PSD, 0–100 Hz zoom.



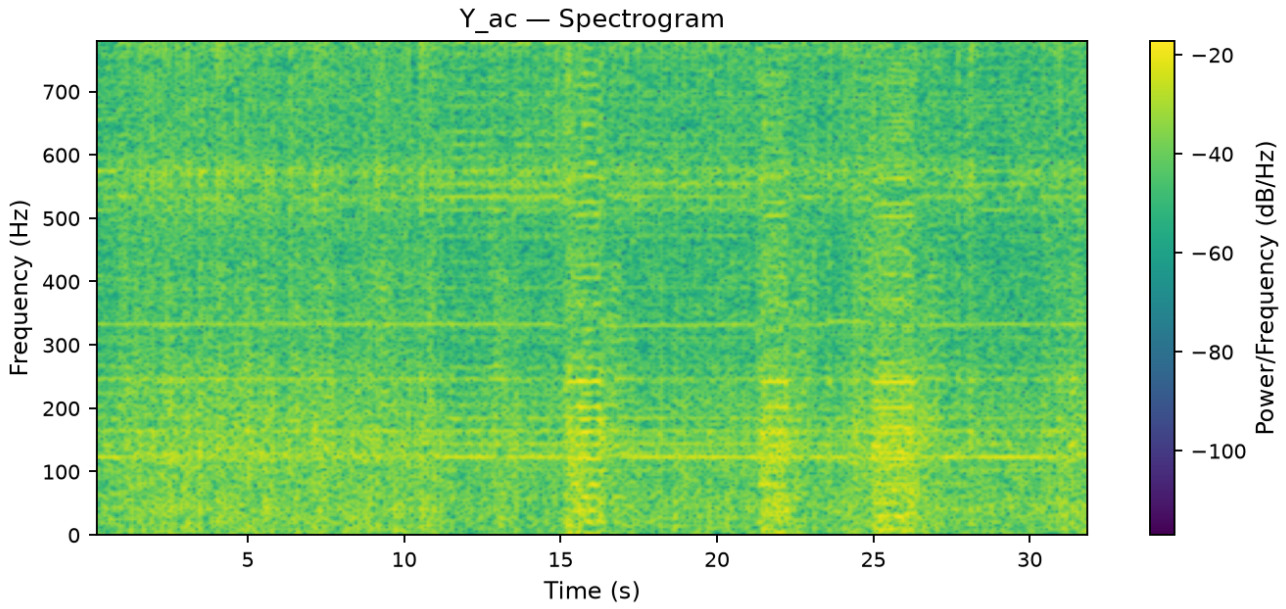
R_g — PSD, 0–100 Hz zoom.

9. Spectrogram Analysis

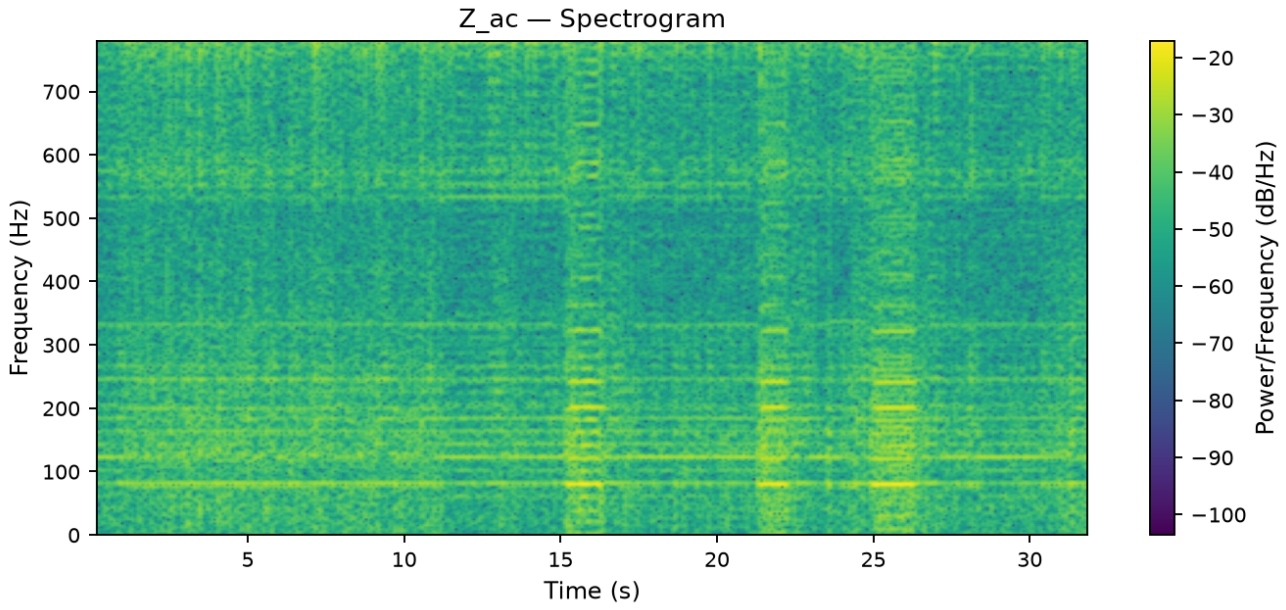
Short-time Fourier spectrograms (nperseg=512, noverlap=384) over the complete available frequency range.



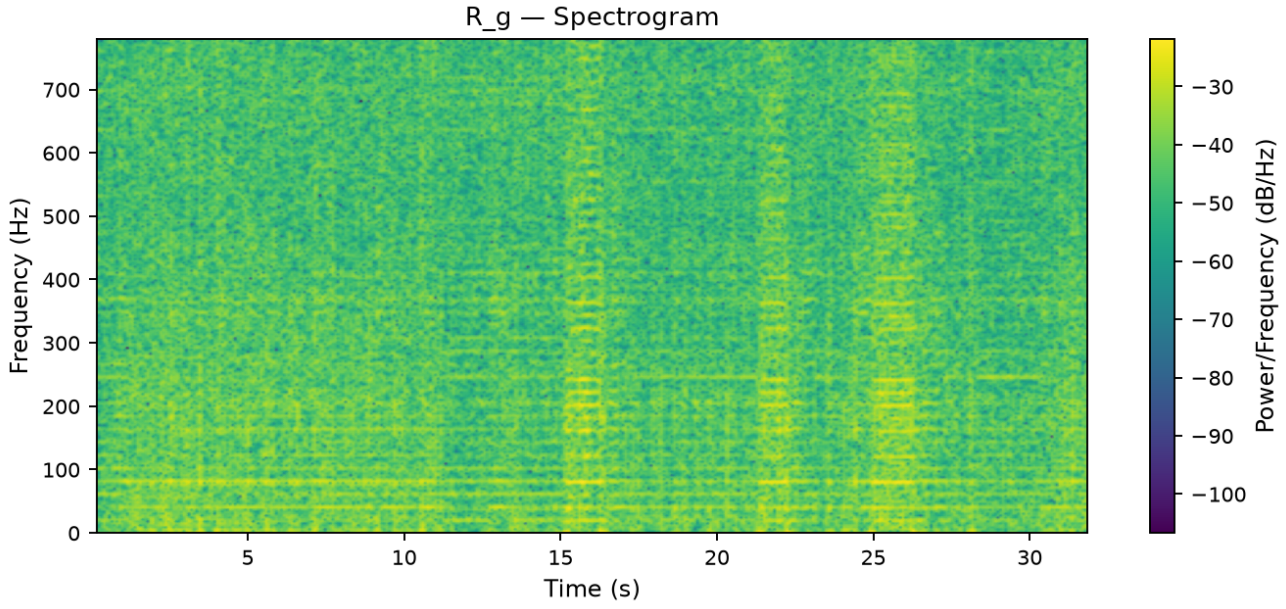
X_ac — spectrogram.



Y_ac — spectrogram.



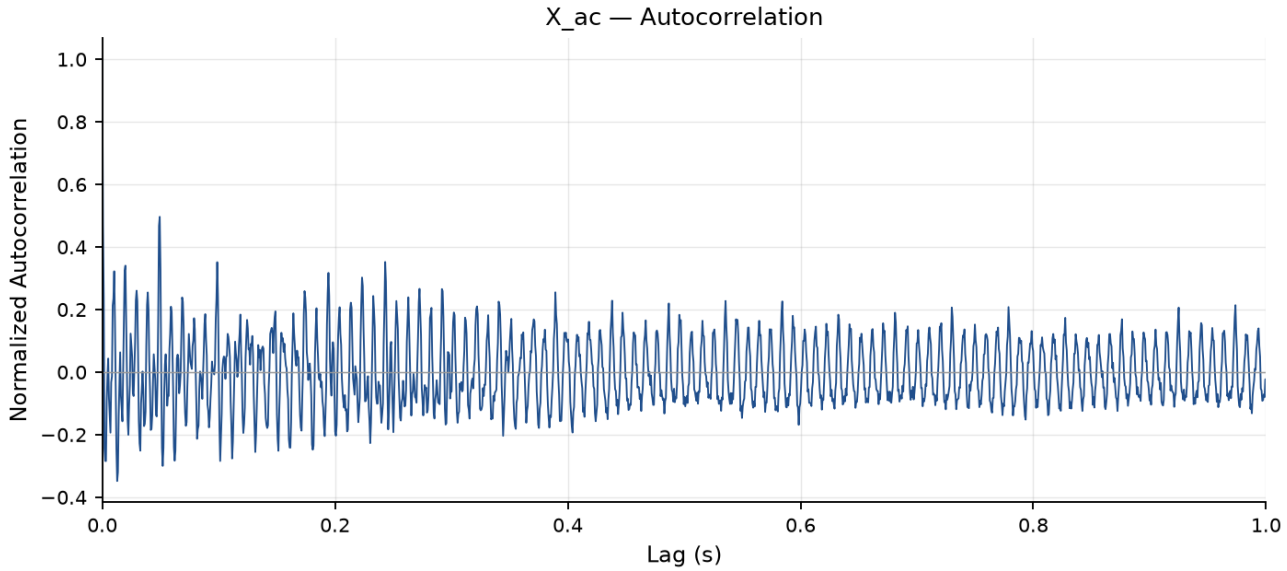
Z_ac — spectrogram.



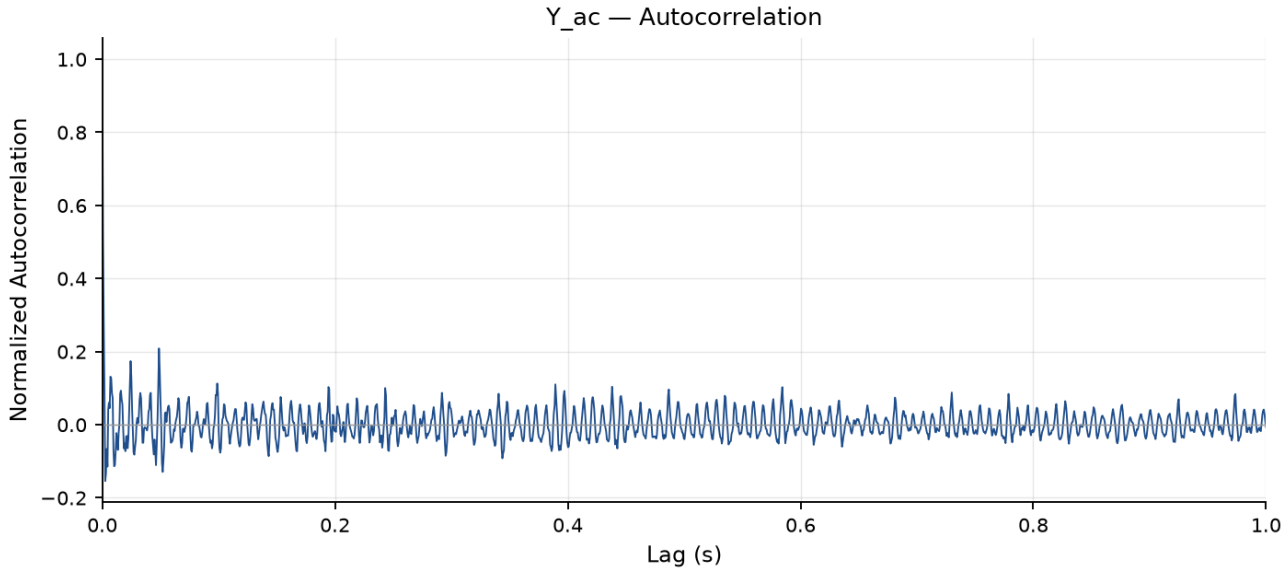
R_g — spectrogram.

10. Autocorrelation Analysis

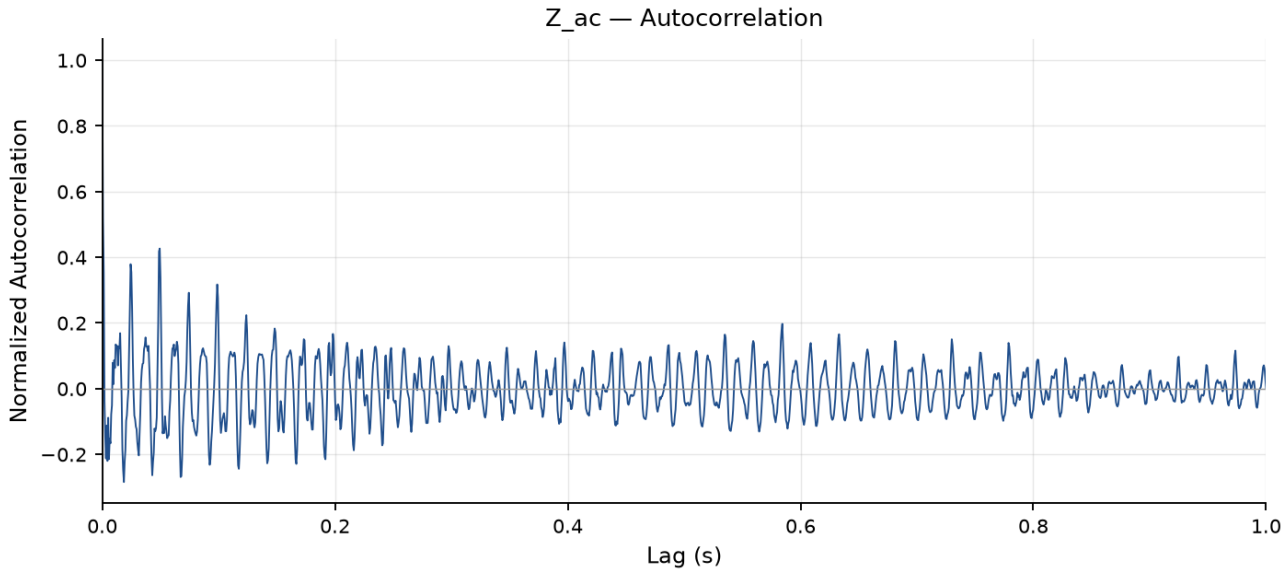
Normalized autocorrelation up to a 1.0 s lag, for visual inspection of periodicity.



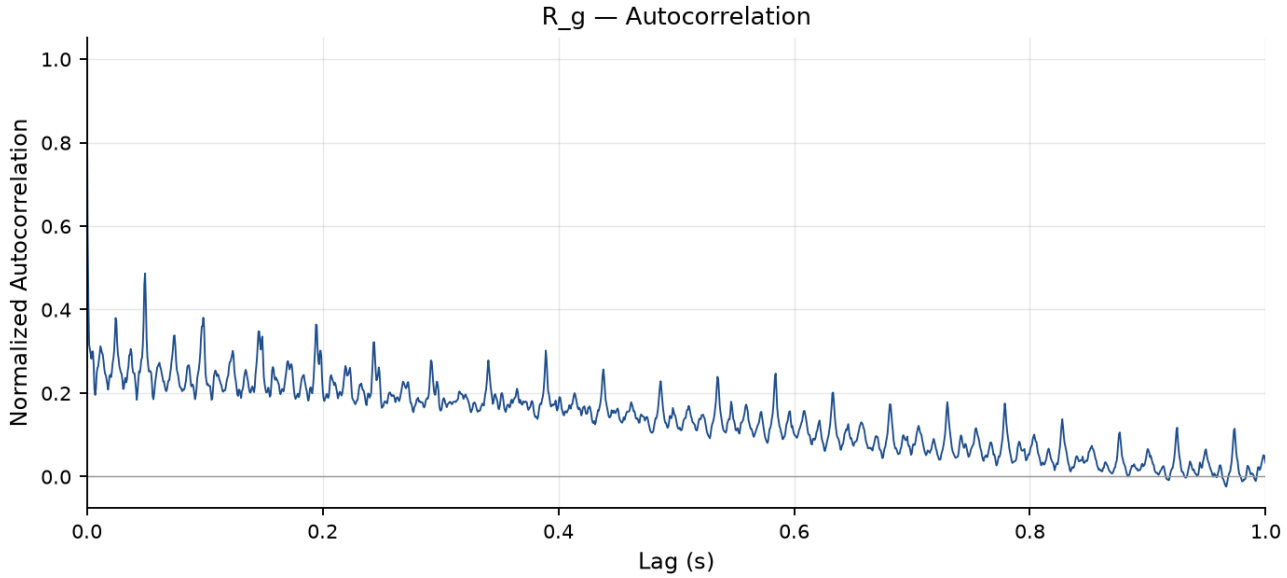
X_ac — autocorrelation.



Y_ac — autocorrelation.



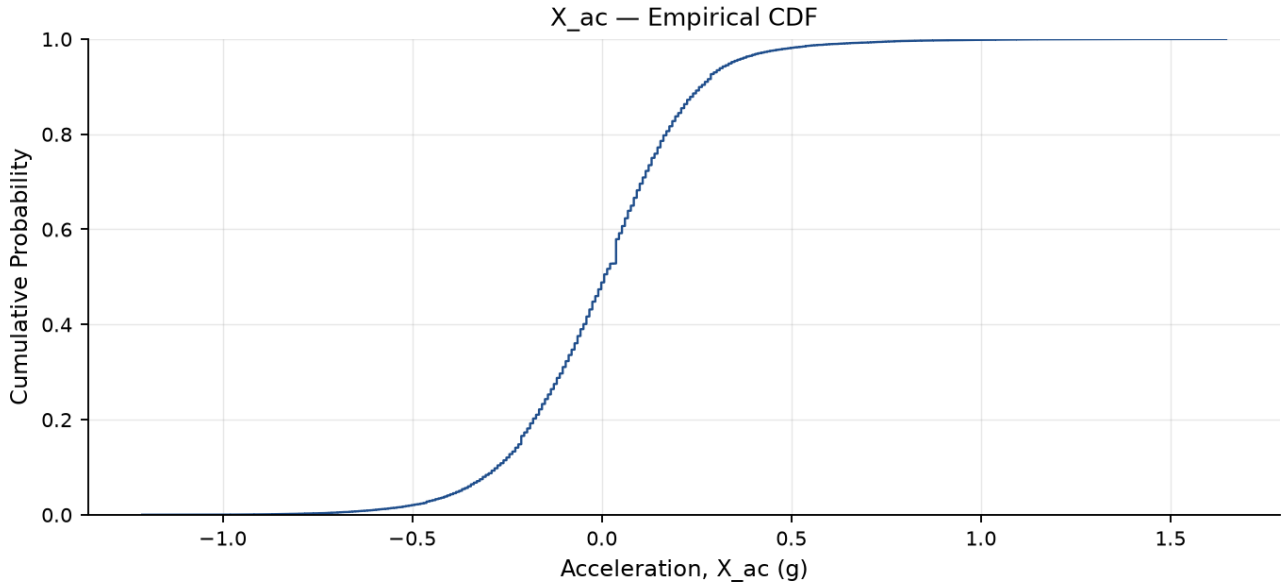
Z_ac — autocorrelation.



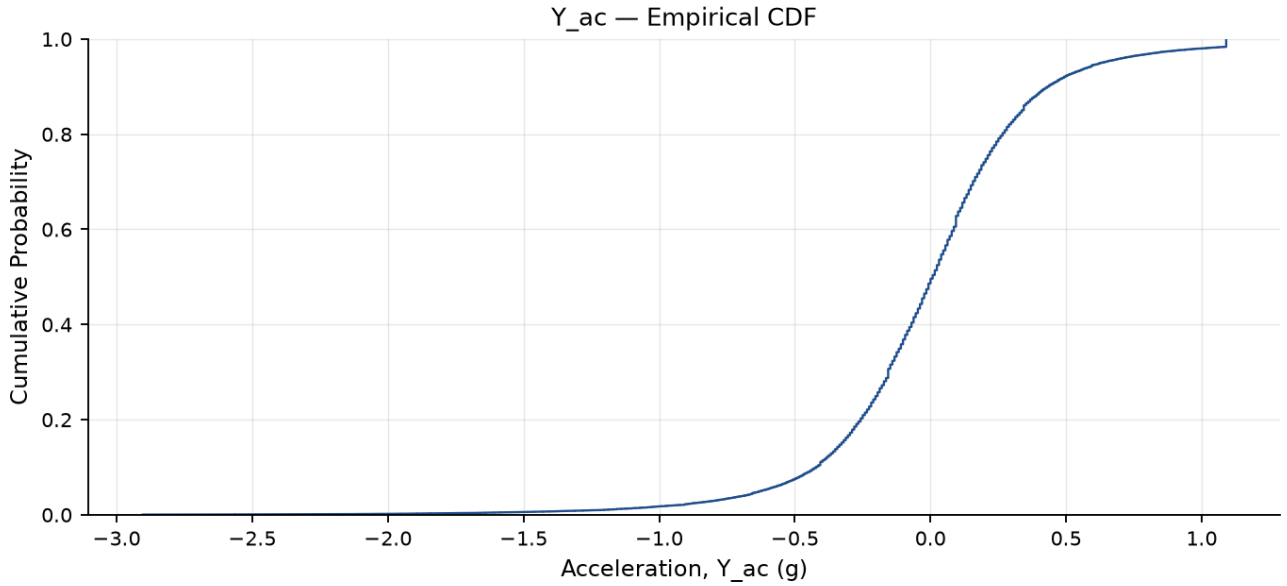
R_g — autocorrelation.

11. Empirical CDF

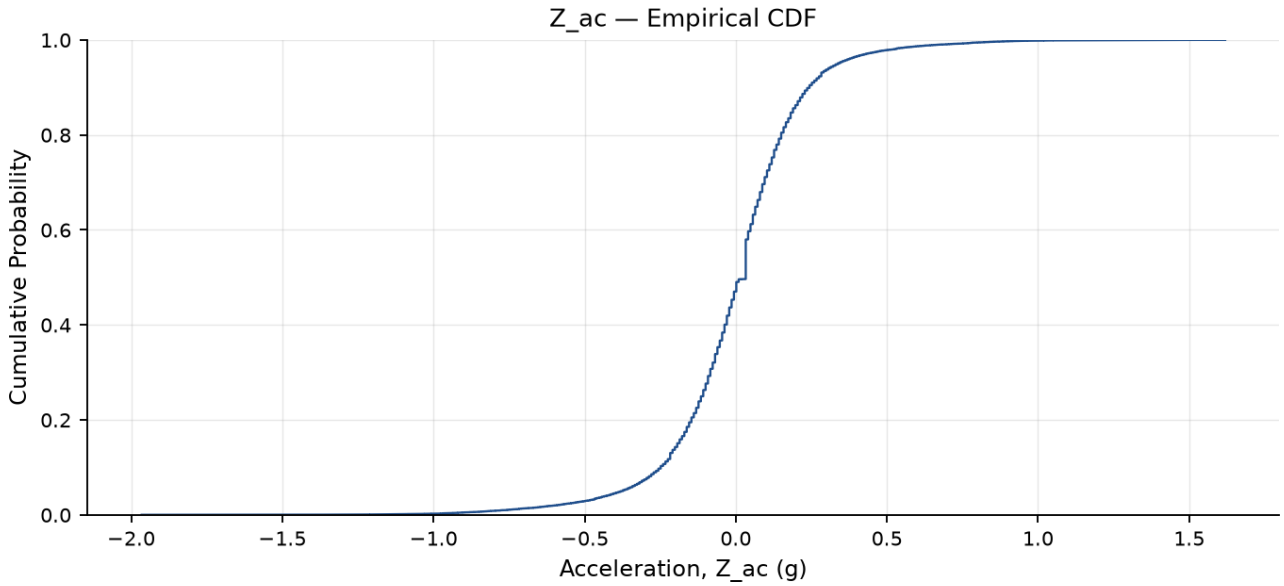
Empirical cumulative distribution functions computed directly from the measured data (no assumed distribution fitted).



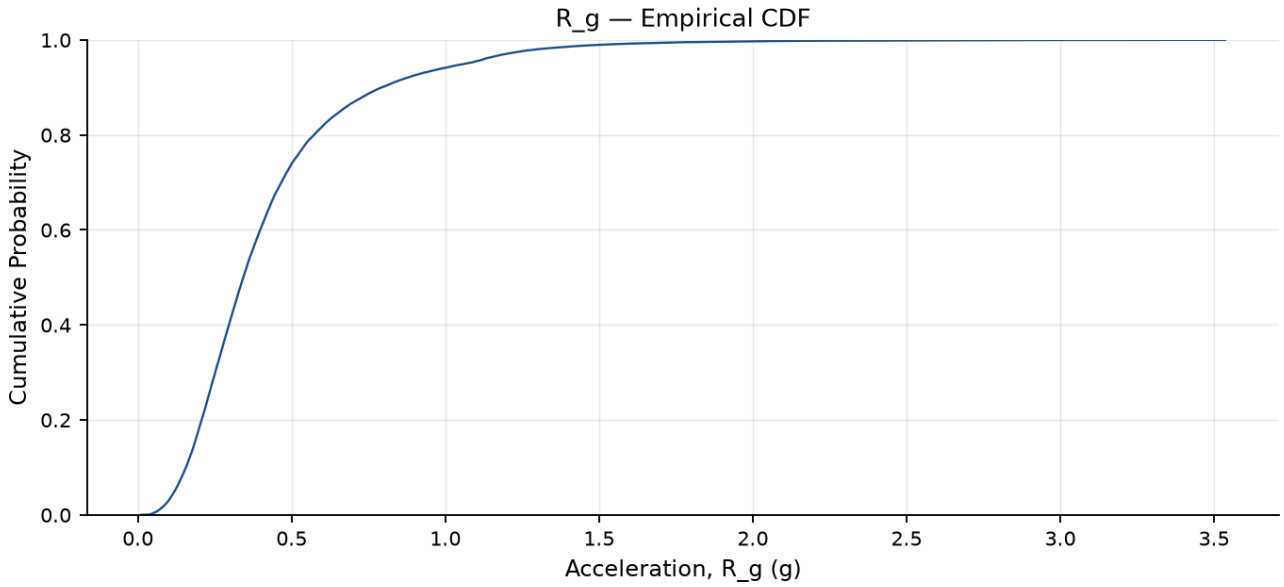
X_ac — empirical CDF.



Y_ac — empirical CDF.



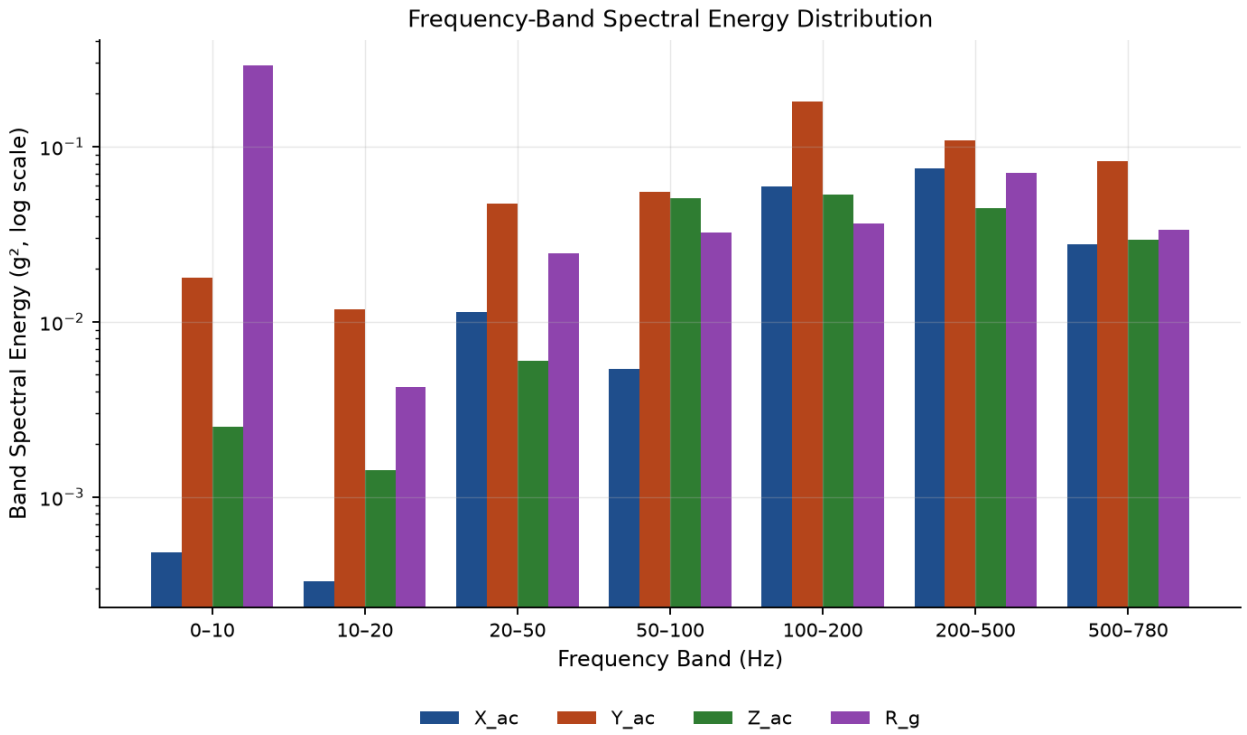
Z_ac — empirical CDF.



R_g — empirical CDF.

12. Spectral Energy

Spectral energy distributed across fixed visualization frequency bands (final band extended to the actual Nyquist frequency, 779.5 Hz). Bands are for visualization only and were not used as an analysis boundary elsewhere.



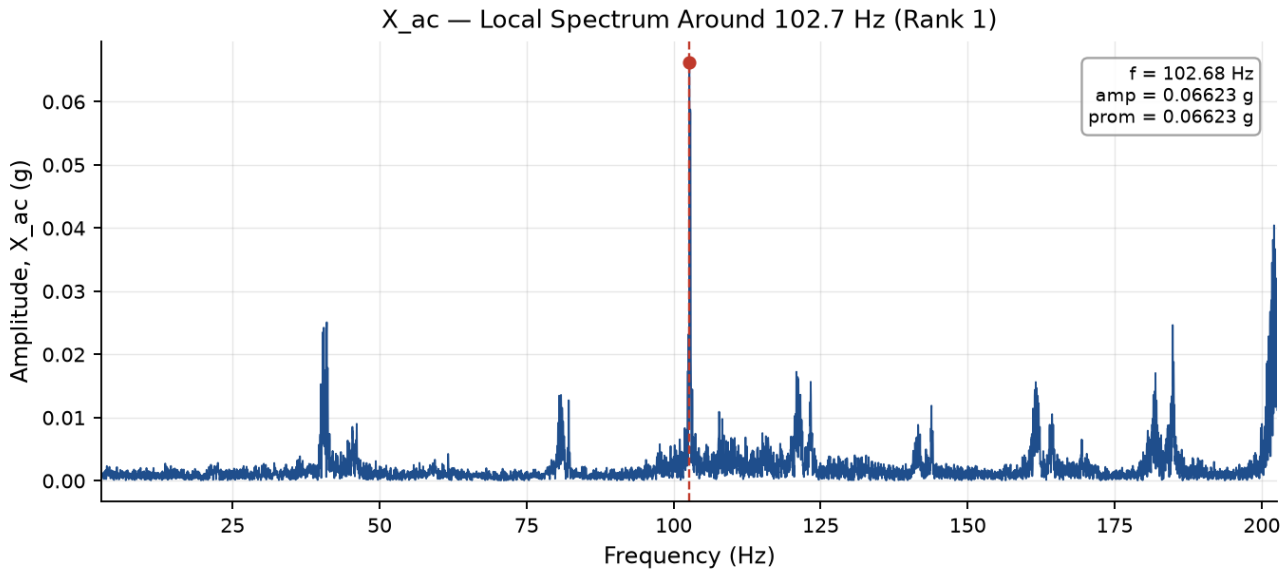
Frequency-band spectral energy, all signals.

Band (Hz)	X_ac	Y_ac	Z_ac	R_g
0–10.0	4.8457e-04	0.017755	0.0025203	0.28984
10–20.0	3.2880e-04	0.011772	0.0014178	0.0042512
20–50.0	0.011323	0.047262	0.0060147	0.024525
50–100.0	0.0053703	0.054892	0.050788	0.03243
100–200.0	0.059135	0.18096	0.052977	0.03655
200–500.0	0.074687	0.10886	0.044721	0.070693
500–779.5	0.027517	0.082531	0.029266	0.033493

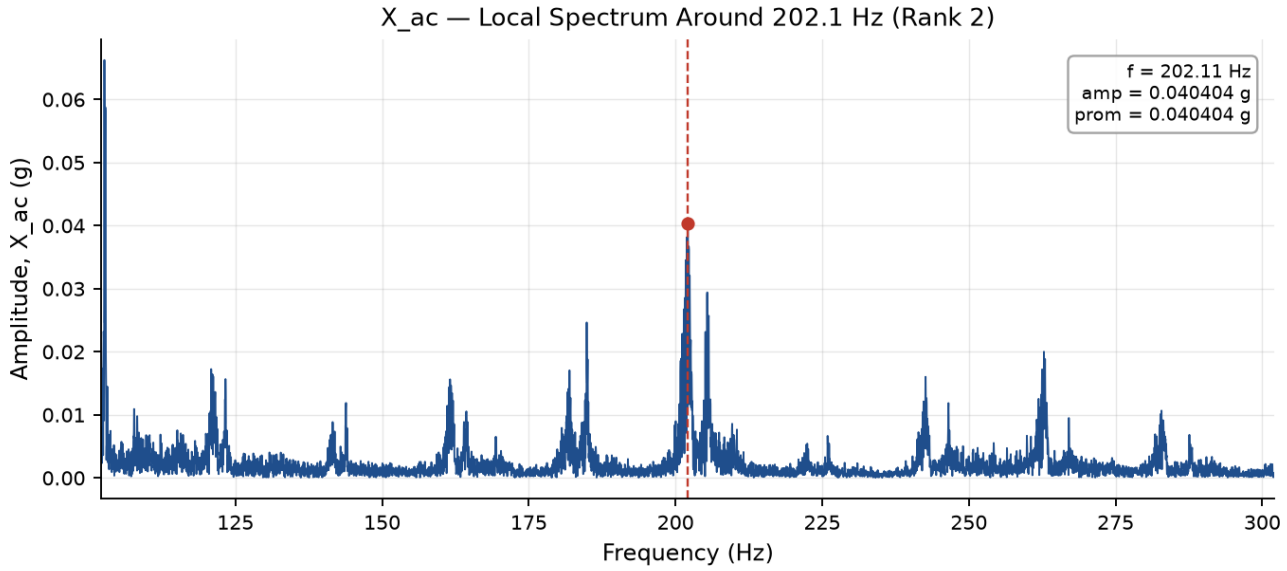
13. Local Spectral Analysis (Automatic ±100 Hz Zoom)

For the strongest automatically detected significant peaks in each signal's full spectrum (adaptive prominence, see Section 6/methodology), a local ±100 Hz window is shown with a compact annotation (frequency, amplitude, prominence). Peaks are ranked by prominence; up to 5 peaks are shown per signal. "Equivalent RPM" values, where given, denote frequency × 60 and describe only what would follow IF that component corresponded to 1× shaft rotation — no such correspondence is asserted by this report. Detailed peak data for all signals is saved to spectral_peaks.csv and local_spectral_analysis.csv for further analysis.

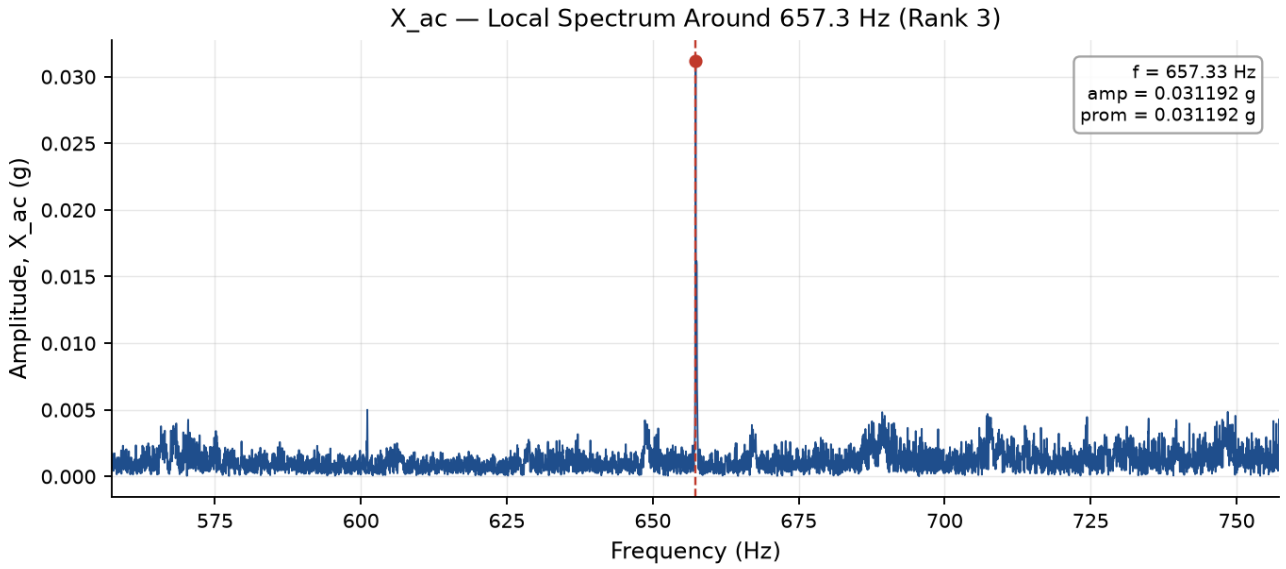
X_ac — Significant Peaks



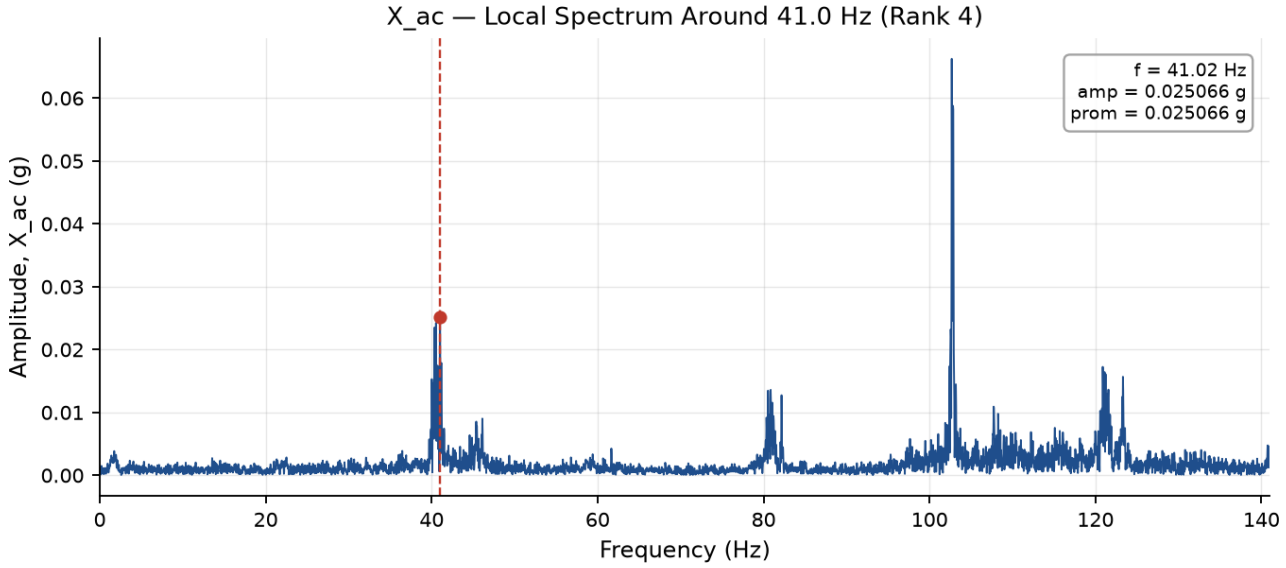
X_ac rank 1: f = 102.68 Hz, amplitude = 0.06623 g, prominence = 0.06623 g (equivalent RPM if 1×: 6161).



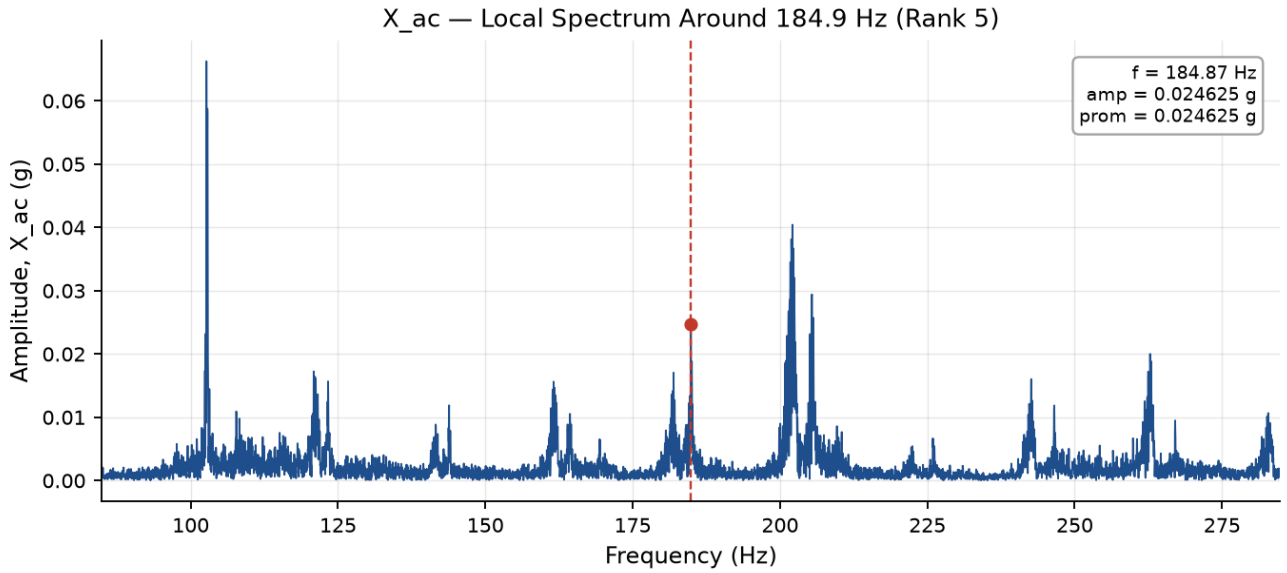
X_ac rank 2: f = 202.11 Hz, amplitude = 0.0404 g, prominence = 0.0404 g (equivalent RPM if 1×: 12126).



X_ac rank 3: f = 657.33 Hz, amplitude = 0.03119 g, prominence = 0.03119 g (equivalent RPM if 1×: 39440).

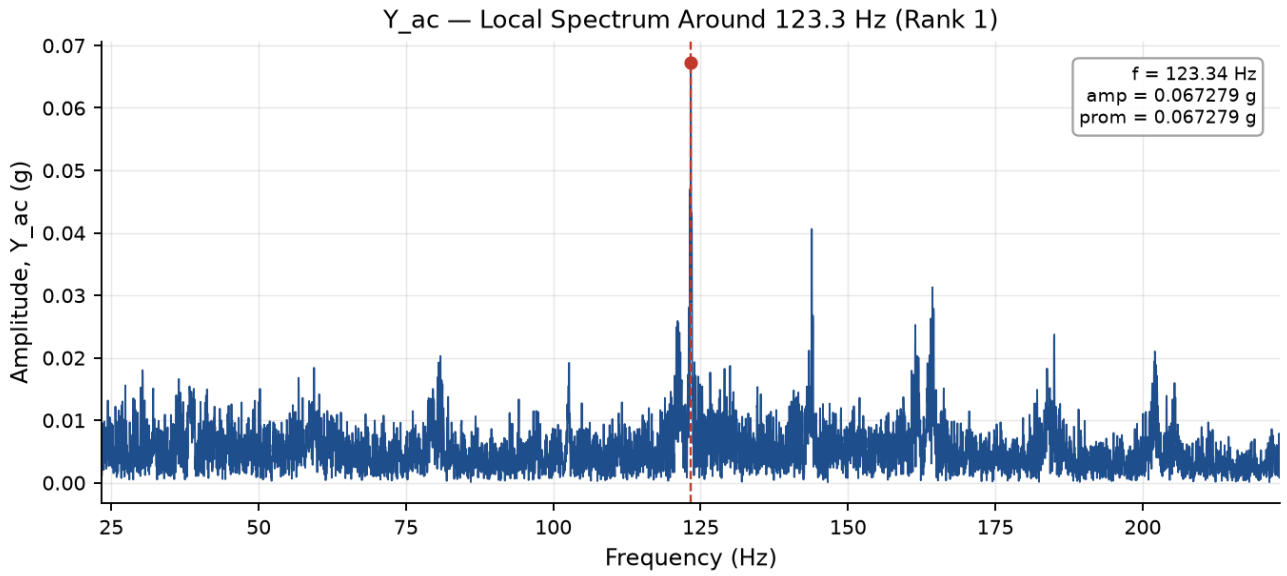


X_ac rank 4: f = 41.02 Hz, amplitude = 0.02507 g, prominence = 0.02507 g (equivalent RPM if 1×: 2461).

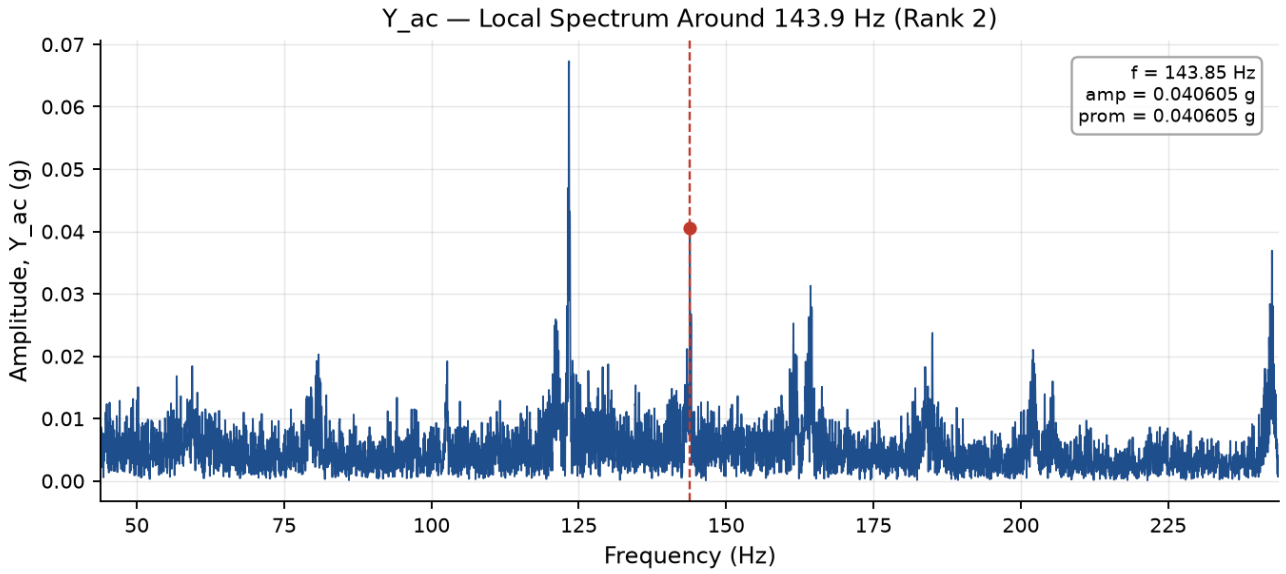


X_ac rank 5: f = 184.87 Hz, amplitude = 0.02462 g, prominence = 0.02462 g (equivalent RPM if 1×: 11092).

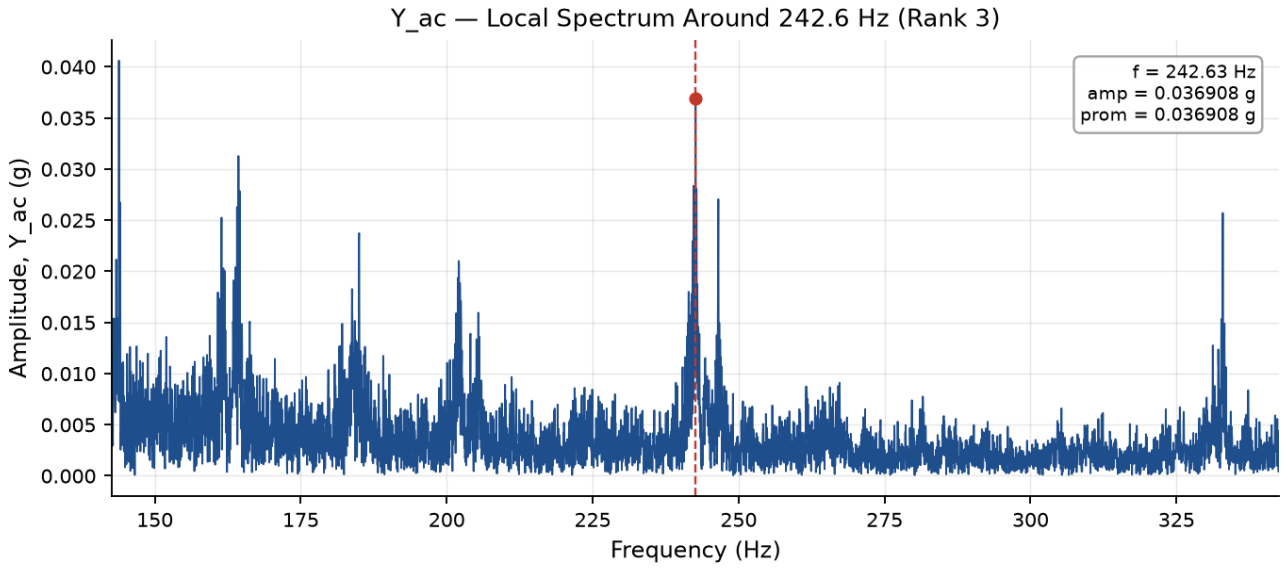
Y_ac — Significant Peaks



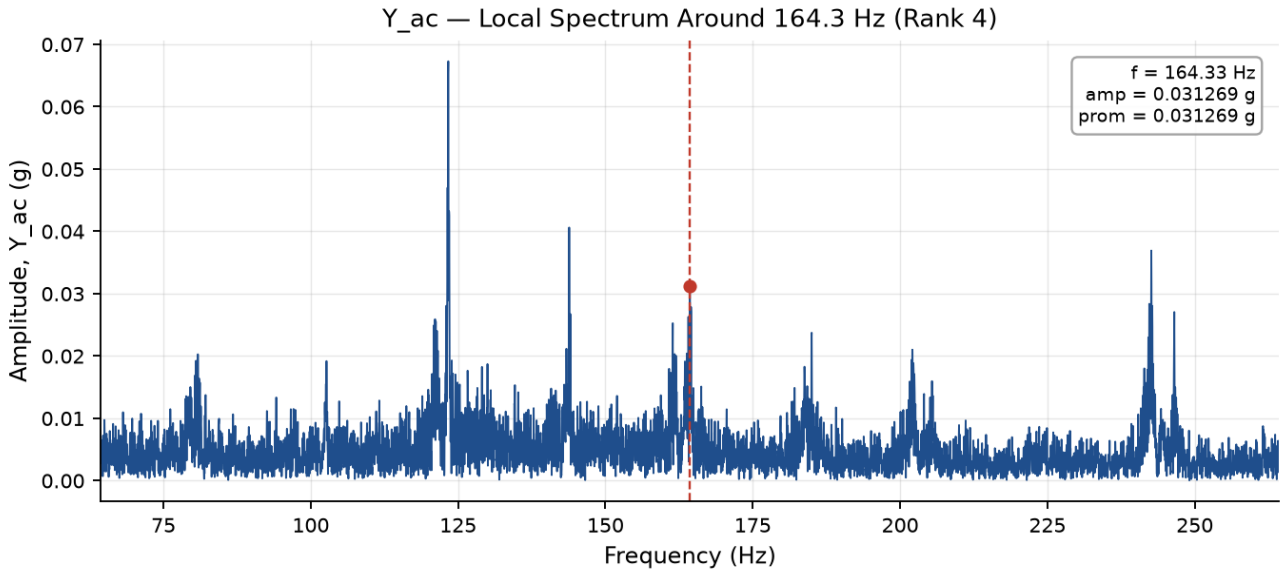
Y_ac rank 1: f = 123.34 Hz, amplitude = 0.06728 g, prominence = 0.06728 g (equivalent RPM if 1×: 7401).



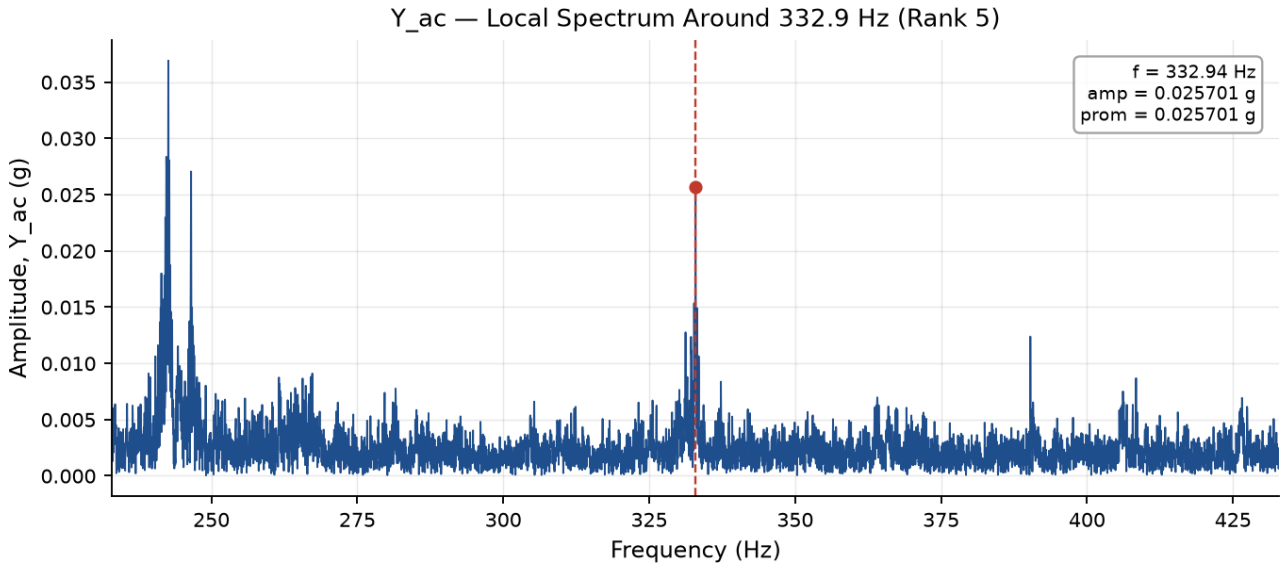
Y_ac rank 2: f = 143.85 Hz, amplitude = 0.0406 g, prominence = 0.0406 g (equivalent RPM if 1×: 8631).



Y_ac rank 3: f = 242.63 Hz, amplitude = 0.03691 g, prominence = 0.03691 g (equivalent RPM if 1×: 14558).

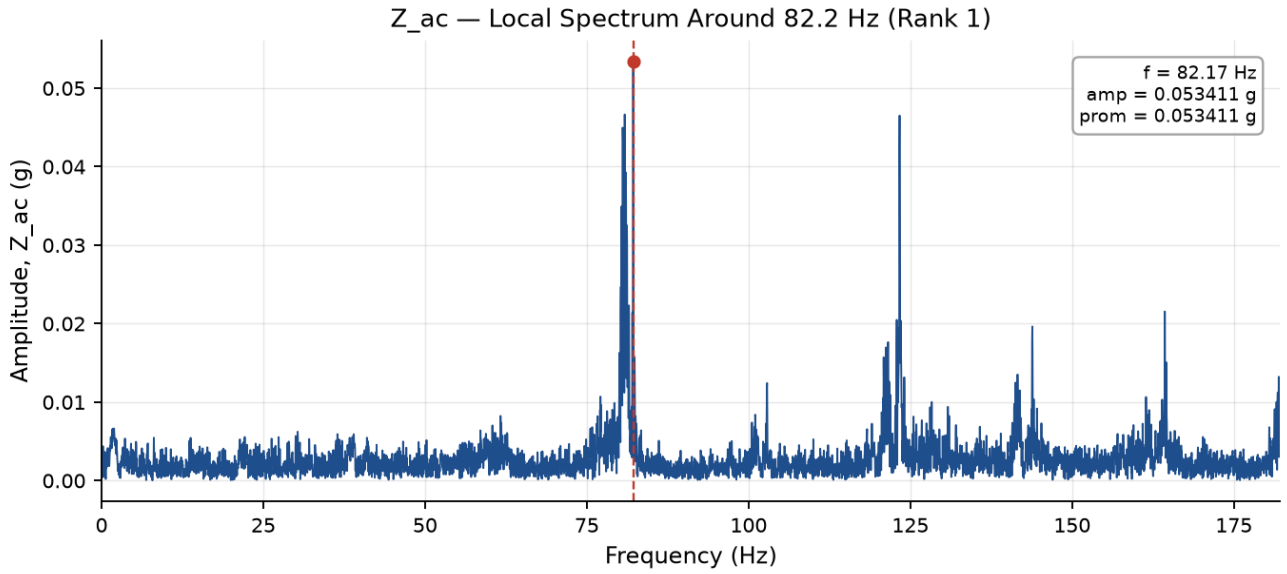


Y_ac rank 4: f = 164.33 Hz, amplitude = 0.03127 g, prominence = 0.03127 g (equivalent RPM if 1×: 9860).

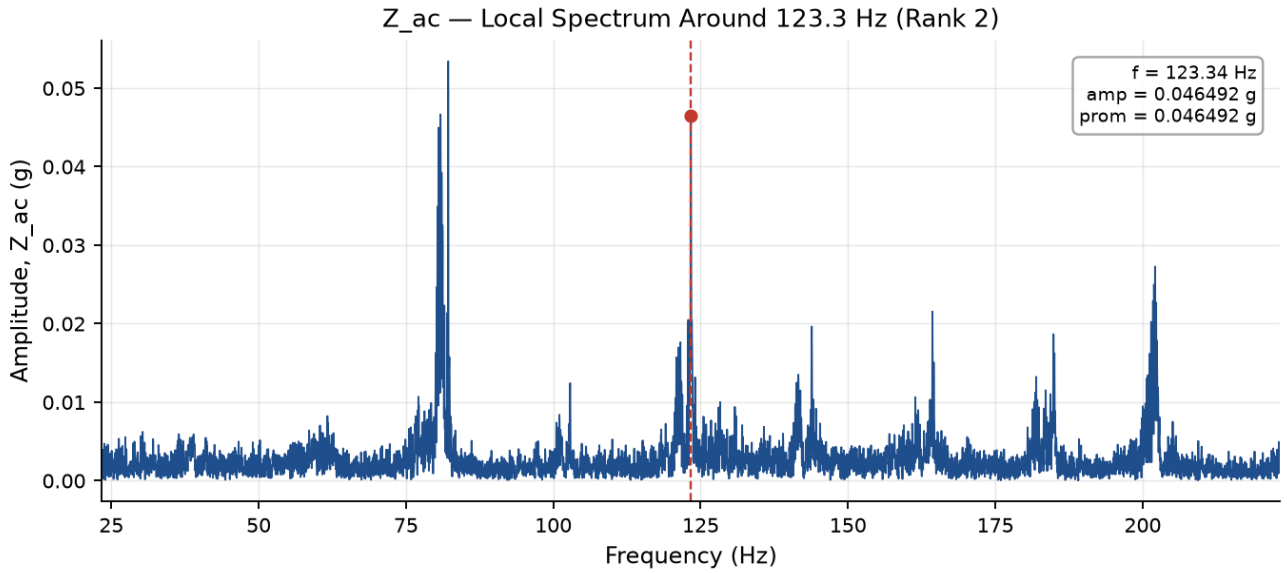


Y_ac rank 5: f = 332.94 Hz, amplitude = 0.0257 g, prominence = 0.0257 g (equivalent RPM if 1×: 19976).

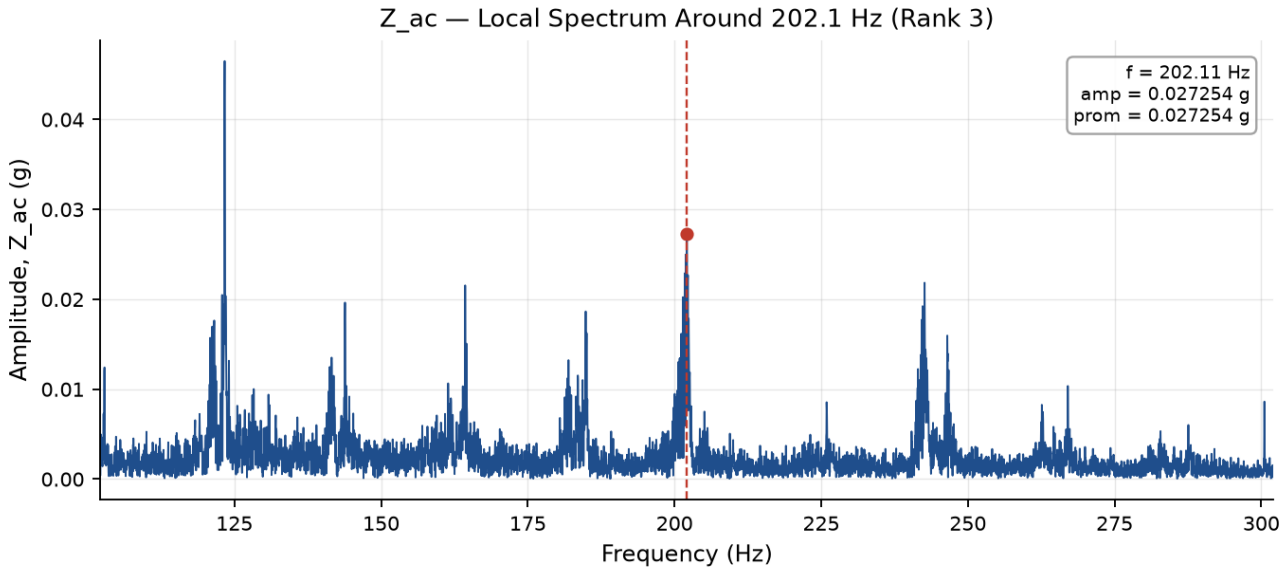
Z_ac — Significant Peaks



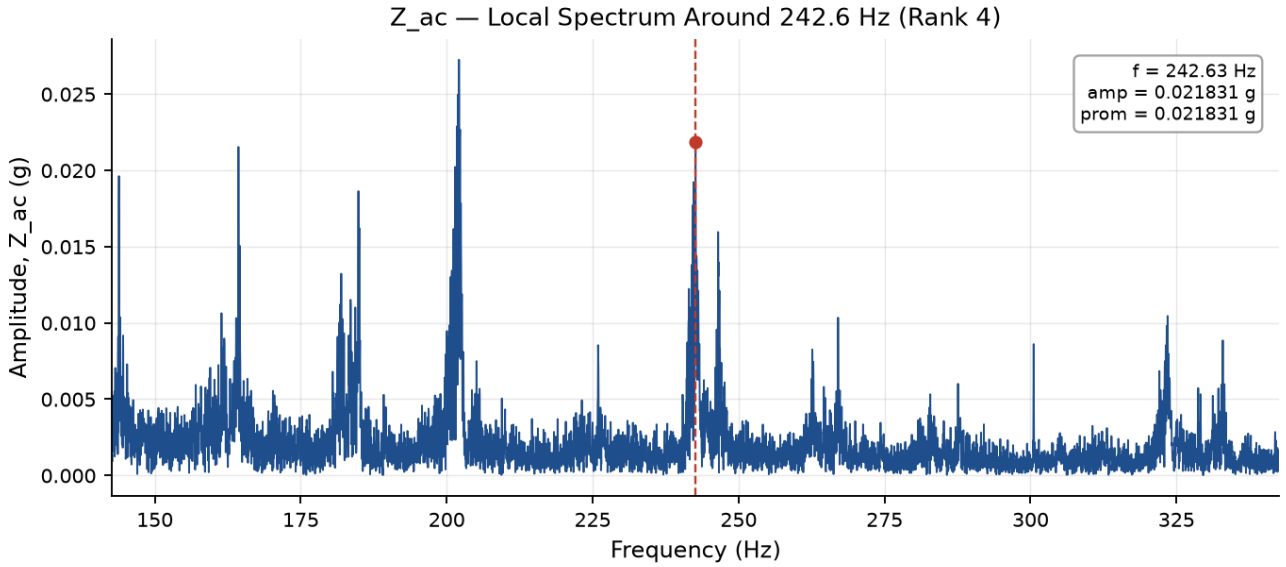
Z_ac rank 1: f = 82.17 Hz, amplitude = 0.05341 g, prominence = 0.05341 g (equivalent RPM if 1×: 4930).



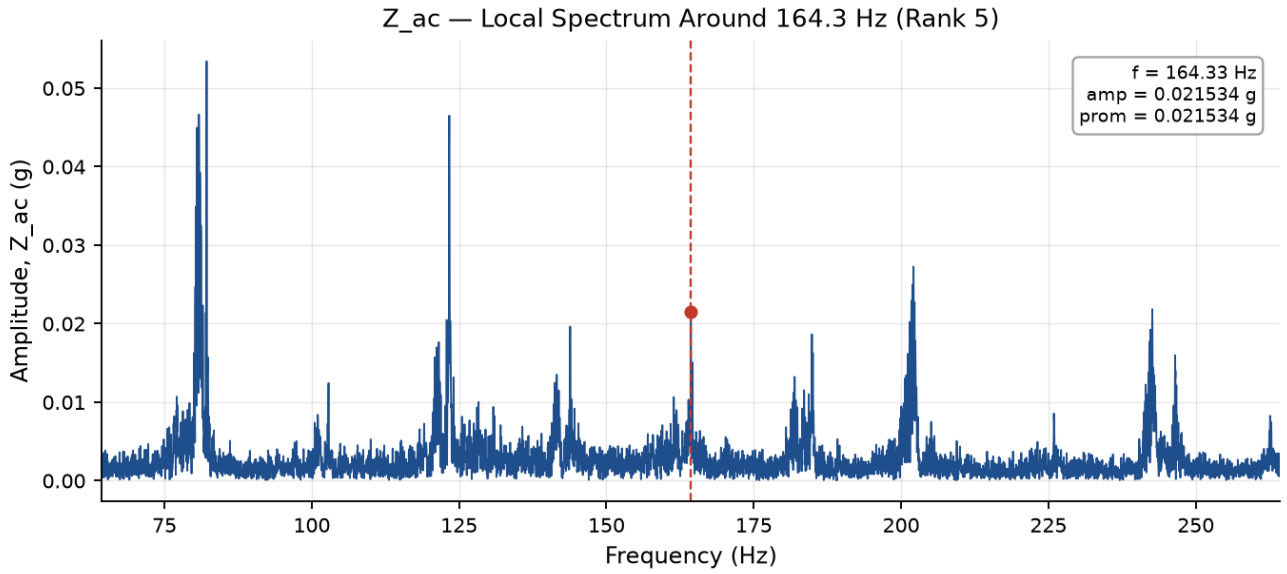
Z_ac rank 2: f = 123.34 Hz, amplitude = 0.04649 g, prominence = 0.04649 g (equivalent RPM if 1×: 7401).



Z_ac rank 3: f = 202.11 Hz, amplitude = 0.02725 g, prominence = 0.02725 g (equivalent RPM if 1×: 12126).

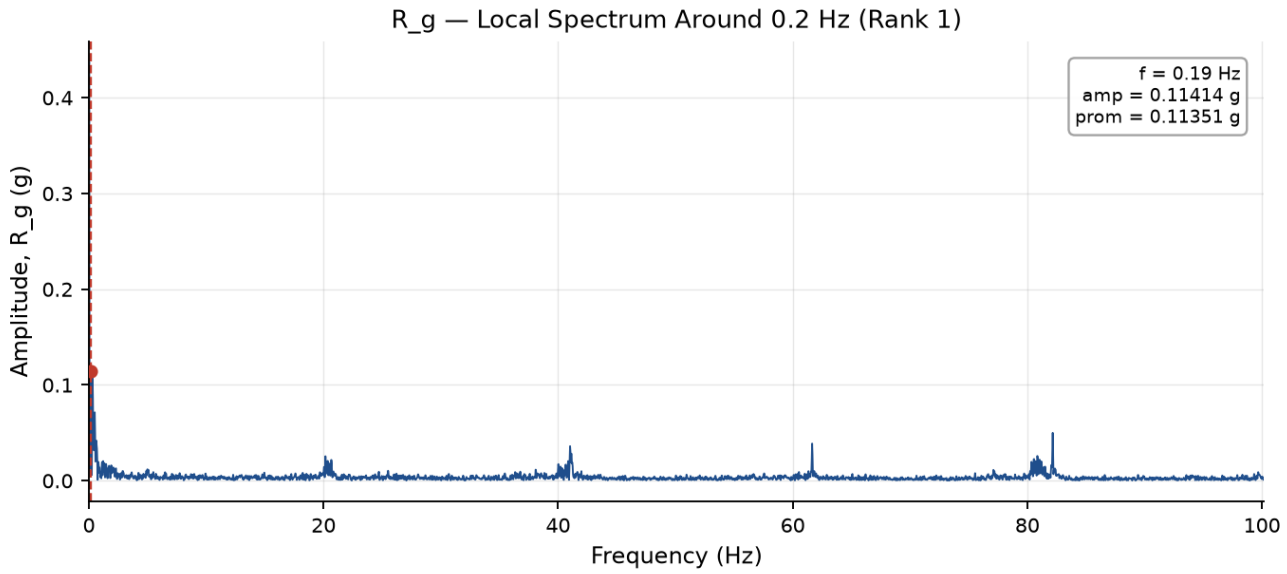


Z_ac rank 4: f = 242.63 Hz, amplitude = 0.02183 g, prominence = 0.02183 g (equivalent RPM if 1×: 14558).

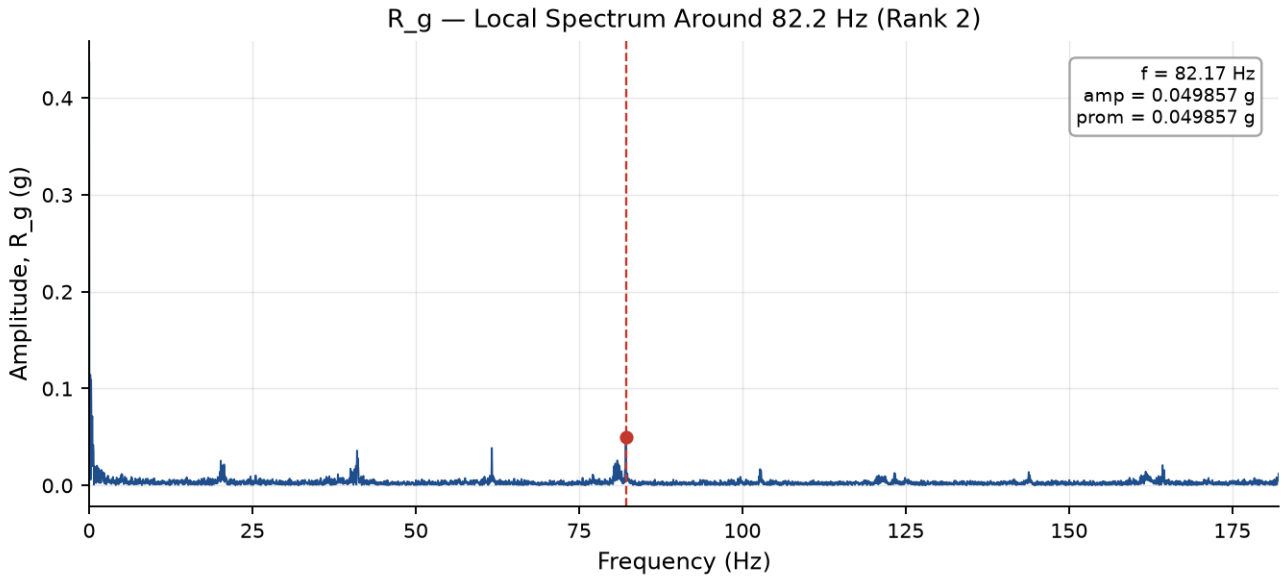


Z_ac rank 5: f = 164.33 Hz, amplitude = 0.02153 g, prominence = 0.02153 g (equivalent RPM if 1×: 9860).

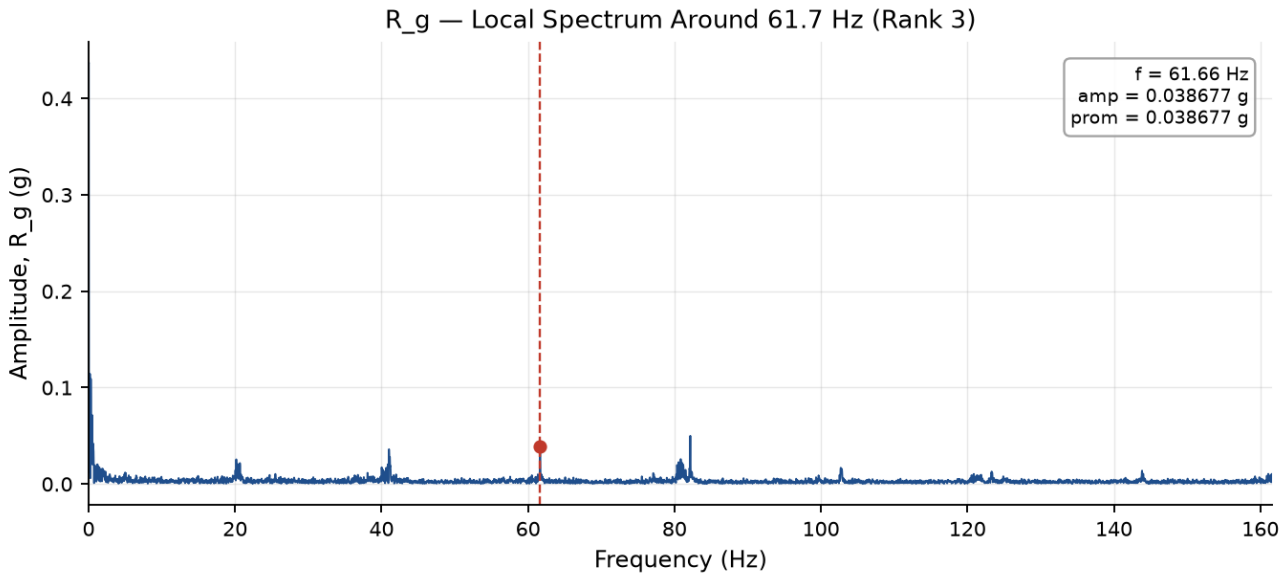
R_g — Significant Peaks



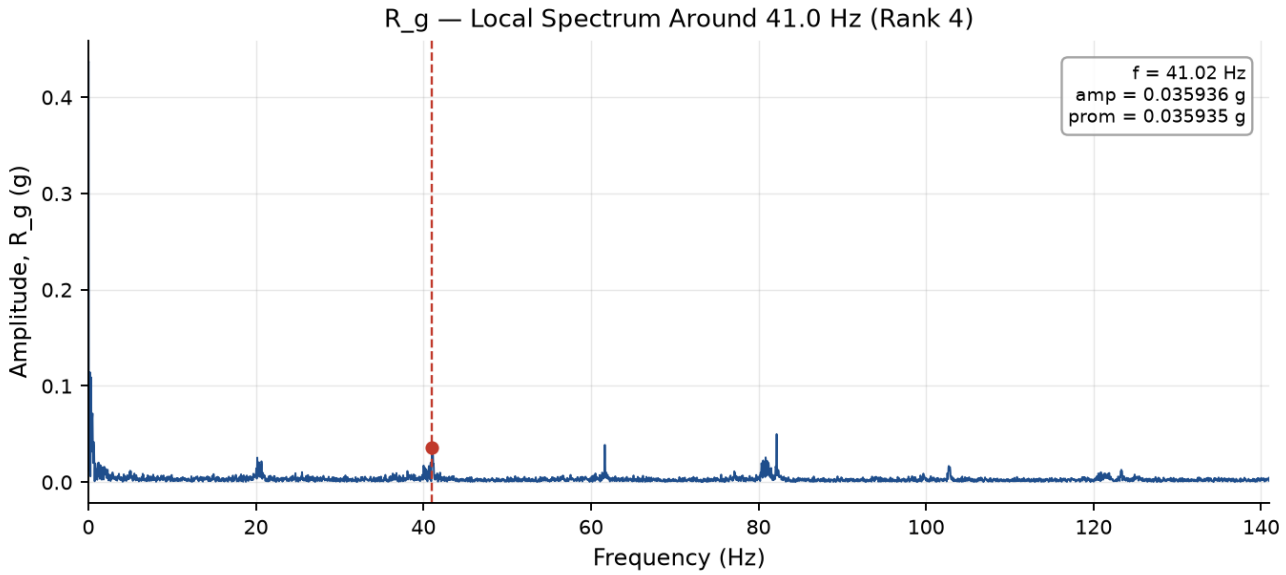
R_g rank 1: f = 0.19 Hz, amplitude = 0.1141 g, prominence = 0.1135 g (equivalent RPM if 1×: 11).



R_g rank 2: f = 82.17 Hz, amplitude = 0.04986 g, prominence = 0.04986 g (equivalent RPM if 1×: 4930).



R_g rank 3: f = 61.66 Hz, amplitude = 0.03868 g, prominence = 0.03868 g (equivalent RPM if 1×: 3699).



R_g rank 4: f = 41.02 Hz, amplitude = 0.03594 g, prominence = 0.03594 g (equivalent RPM if 1×: 2461).